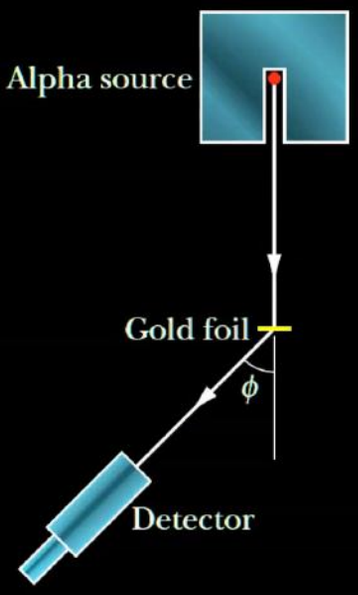
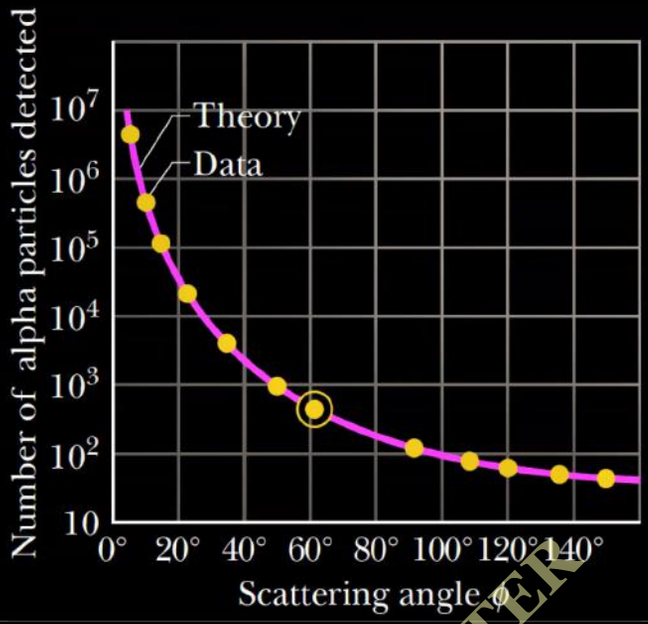
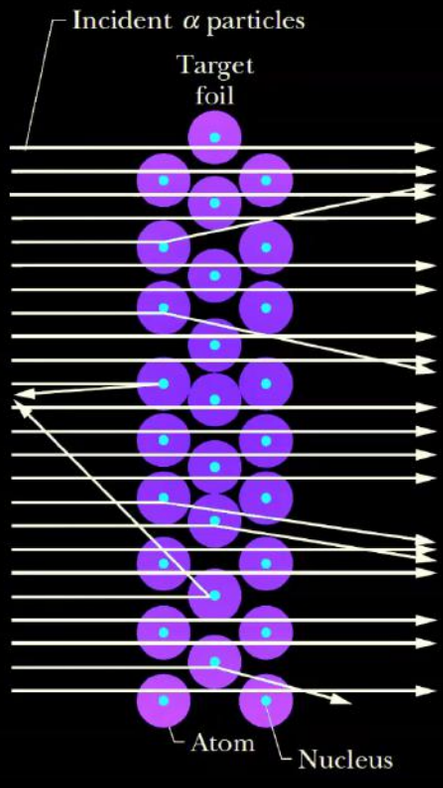
Discovery Of The Nucleus:





The first experimental evidence for presence of nucleus came from Rutherford's alpha particle scattering experiment. It was discussed in some detail during our study of electrostatic potential and also atomic physics. It was found that most alpha particles passed through the Goldfoil undeviated and there were very few (though finite) number of particles which had very large deviations. Such large deviations could only be explained by presence of concentrated positive charge in a very small region of space. The atom is largely hollow space with almost all the mass concentrated in a tiny region called nucleus with dimension of the order of 10^{-15} m or fm (read femtometers). If the nucleus were to be imagined to be the size of a football, the electrons are then orbiting outside the stadium. The atoms are that "hollow".



Example: A 5.3 MeV alpha particle moves head-on towards a Gold nucleus ($Z = 79$). How close does the alpha particle get to the nucleus?

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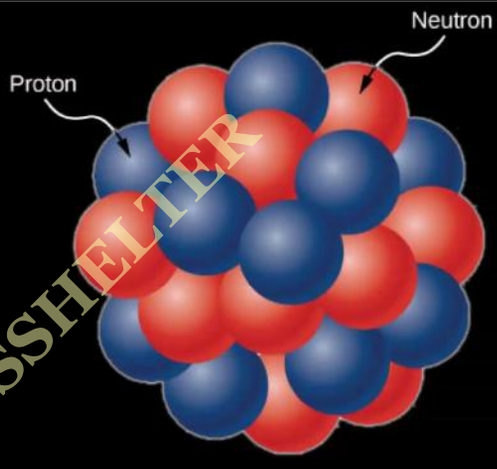
Solution: The kinetic Energy must be converted to electrostatic potential energy at the distance of closest approach. Thus,

$$KE = \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{r} \Rightarrow r = \frac{1}{4\pi\varepsilon_0}\frac{2Ze^2}{KE}$$

$$= \frac{9\times10^9 \times 2\times79\times e\times1.6\times10^{-19}}{5.3\times10^6\, e} \text{ metres} \approx 4.29\times10^{-14} \text{ metres}$$

$$\approx 43 \text{ fm.}$$

SOME NUCLEAR TERMINOLOGY:



Nucleus contains protons (Discovery: Rutherford) and neutrons (discovery: Chadwick). The number of protons in a nucleus is called its atomic number and is represented by symbol Z. The number of neutrons is represented by symbol N. Total number of nucleons (i.e protons & neutrons) is called mass number A thus $A = Z + N$.

ISOTOPES: Nuclei with same Z but different A (or N) such as $^{6}_{3}Li$ and $^{7}_{3}Li$, or $^{1}_{1}H$, $^{2}_{1}H$ and $^{3}_{1}H$.

ISOTONES: Nuclei with same neutron numbers. Such as: $^{37}_{17}Cl$ and $^{39}_{19}K$ or $^{17}_{7}N$, $^{18}_{8}O$ and $^{19}_{9}F$.

ISOBARS: Nuclei with same mass numbers. e.g: $\left[^{40}_{18}Ar, {}^{40}_{20}Ca\right]$ and $\left[^{76}_{32}Ge, {}^{76}_{34}Se\right]$

ISOMERS: Nuclei with same Z and N but slightly different properties due to internal arrangement in the nucleus (often only one isomer is stable, others are either metastable or unstable) e.g. $^{80}_{35}Br$ and $^{80}_{35}Br^*$ (Astenish is used to mean unstable state).

ISODIAPHERS: Nuclei with same N−Z (or A−2Z) i.e same excess of neutrons over protons are called isodiaphers e.g: $^{23}_{11}Na$ and $^{27}_{13}Al$.

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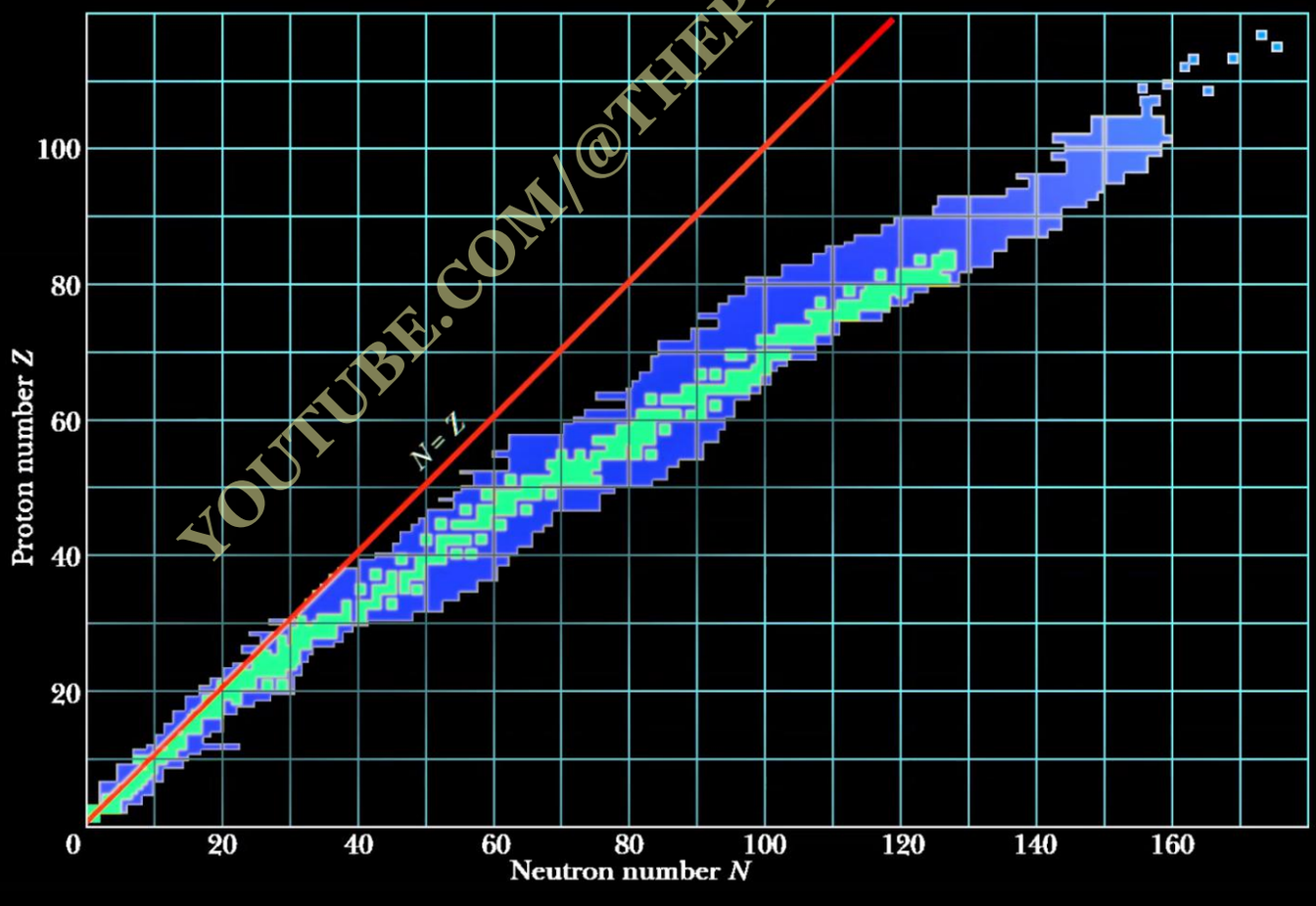
DECAY OR DISINTEGRATION: Many istotopes of elements are unstable and convert to another nucleus by emitting a particle. Such a nuclear reaction is called disintegration (or decay). For example Au has 32 istopes out of which 31 are unstable. (Ranging from ^{173}Au to ^{204}Au). In these only ^{197}Au is stable and others are radioactive (undergo decay) e.g.:

$$^{198}_{79}Au \longrightarrow {}^{198}_{80}Hg + {}^{0}_{-1}e + \bar{\nu}$$

Here $\bar{\nu}$ is called antineutrino which has zero charge but carries some energy.

→ In every nuclear reaction, mass number, charge, momentum and spin are conserved. (Also mass energy).

NUCLIDIC CHART:



A plot of proton number (Z) vs neutron number (N) for various nuclei found in nature or made in lab is called a nuclidic chart. In this chart the stable nuclides are shown in green.

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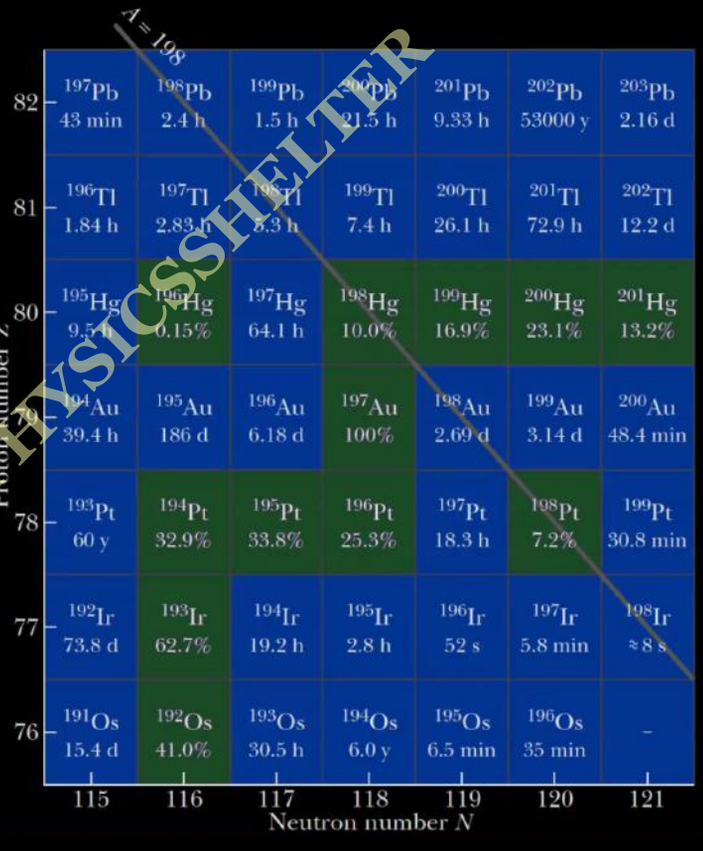
These shown in blue colon are unstable and undergo nuclear decay. It can be seen that for lighter elements N and Z are nearly equal, while for heavier nuclei N exceeds the proton number significantly. (For instance ^{197}Au has 118 neutron but only 79 protons). We will understand its reason in an upcoming section.

→ There are no stable nuclei beyond Z > 83 (Bismuth).

→ No elements with Z > 92 occurs naturally (though some have been made in laboratories).

→ All naturally occuring radioactive heavy elements (Z > 83) decay to one of the isotopes of Lead or Bismuth.

→ The adjacent figure shows zoomed in version of Nuclidic chart centred on ^{197}Au. The elements in green are stable (relative abundance is indicated) while those in blue are radioactive (Half lives indicated). (Half live means time required for half the nuclei to decay). The inclined line crosses the isobars. Notice that there are more numbers of stable nuclei with both Z and N even and in the zoomed in table there is no nuclide with both odd. This has something to do with spin pairing. Infact in nature only five odd-odd species are stable namely $^{2}_{1}H$, $^{6}_{3}Li$, $^{10}_{5}B$, $^{14}_{7}N$ and $^{180}_{73}Ta$.



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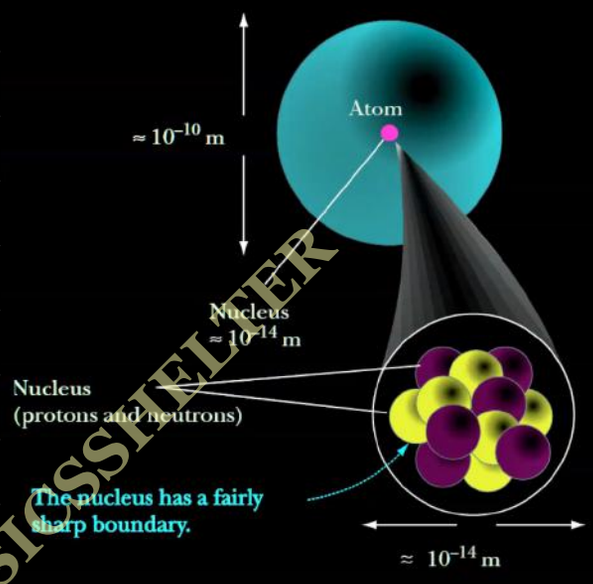
NUCLEAR RADII: Nuclei may be approximately assumed to be spherical (though some are notably ellipsoidal). Nuclear density can also be assumed to be nearly constant thus volume is directly proportional to mass number accordingly the nuclear radii may be taken as $R = R_0 A^{1/3}$ where $R_0 = 1.2\ fm$

Mass of a proton is about 1.66×10^{-27} kg accordingly we can use it to estimate nuclear density:

$$\rho = \frac{Au}{4/3\,\pi r^3} = \frac{1 \times 1.66 \times 10^{-27}\ kg}{4/3\,\pi\,(1.2 \times 10^{-15})^3} = 2.3 \times 10^{17}\ kg/m^3$$

Thus nuclear density is 2.3×10^{17} kg/m^3.

→ The adjacent figure gives a schematic idea of relative sizes of the atom and the nucleus. Nuclear volume is about 10^{12} times smaller than atomic volume. The matter is almost entirely empty space.



Mass Energy Conversion: A consequence of Einsteins theory of relativity is that mass and energy are equivalent (or interconvertible). This can be proved just using the fact that speed of light remains constant in every frame and some high school mathematics, though we will not be doing the proof here. The mass energy conversion equation is given as $\boxed{E = mc^2}$. In nuclear reactions some mass may get converted to Energy and the energy liberated in the process is given as $\boxed{Q = -\Delta m c^2}$ where Δm is the mass of the products − mass of the reactants in the nuclear reactions.

Atomic Masses And Mass Excess: All atomic masses are standardized with respect to ^{12}C atom. The mass of ^{12}C atom is taken as exactly 12 u. In nuclear reactions the atomic mass numbers are conserved so sometimes it is easier to drop the nearest integer part from the atomic masses. Thus we define the quantity mass excess as: The mass of an atom minus its mass number is called mass excess, mathematically:

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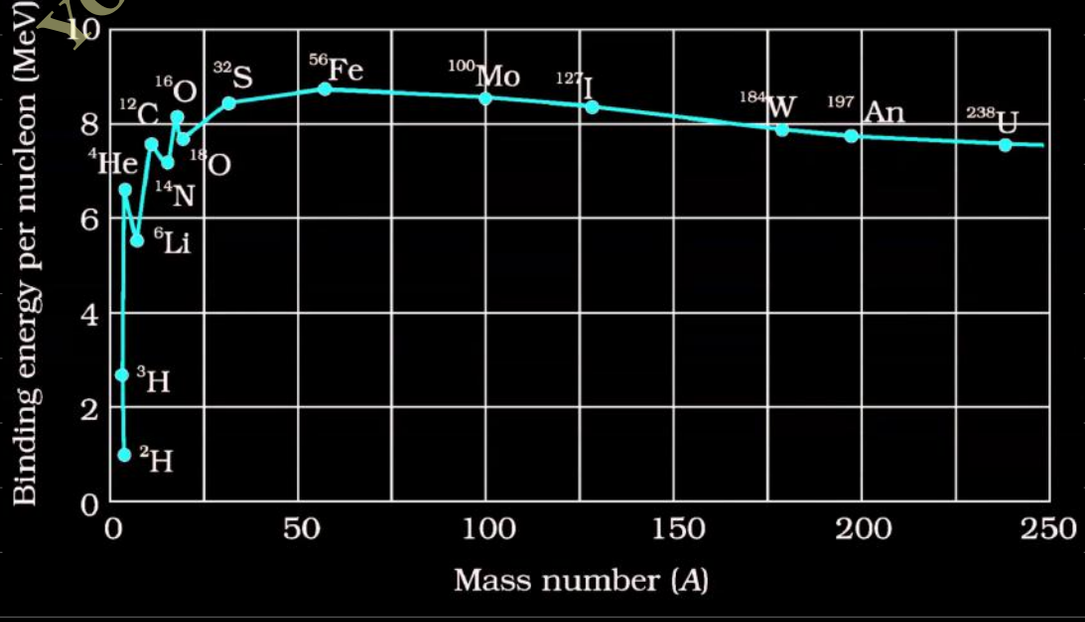
$\Delta = M - A$ where M is the mass of the atom and A is the mass number. For instance mass of ^{197}Au is 196.966552 u thus, its mass excess is $196.966552 - 197 = -0.033448$ u.

Nuclear Binding Energy: Whenever nucleons come together the form a nucleus, the total mass M of the nucleus is less than Σm i.e the summation of masses of the nucleons. This also means mass energy Mc^2 of a nucleus is less than the mass energy Σmc^2 of the all the nucleons. This loss of mass energy in the process of formation of the nucleus is called the binding energy of the nucleus. Thus:

$$B.E. = \Sigma mc^2 - Mc^2$$

, this is also the minimum theoretical energy that is required to split the nucleus in to its component nucleons. More the binding, more the energy required to split it apart completely. A nucleus will be transformed in to another species nuclide even if one of its nucleons is plucked out. Thus, a measure of nuclear stability is binding energy per nucleon, which is simply the ratio of total binding energy to the mass number. Thus $BE_n = \frac{B.E.}{A}$. This is average amount of energy to pluck apart a nucleon (could be slightly different for removing a proton and a neutron).

Binding Energy per Nucleon:



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The above graph from NCERT shows variation of binding energy per nucleon for some elements. The main features worth mentioning in this graph are:

→ Binding Energy per nucleon is nearly constant i.e practically independent of mass number for nuclei with mass numbers between 30 and 170. The curve has maximum of 8.75 MeV for A = 56 (Iron 56) and 7.6 MeV for A = 238 (Some books though state that ^{62}Ni is most strongly bound, but we will stick to NCERT Data).

→ Binding energy per nucleon is low for both light (A < 30) and heavy (A > 170) nuclei

→ Due to low binding energy per nucleon, under suitable conditions, lighter nuclei fuse to form medium sized nuclei and heavy nuclei can fission in to medium sized nuclei thus obtaining a stable configuration.

→ The stability of nuclei implies that nuclear forces must be sufficiently attractive to overcome coulombic repulsion and bind the nucleus together

→ The constancy of binding energy is due to the fact that nuclear forces are very short ranged. Thus practically only immediate neighbors contribute to the binding energy and number of neighbors for a nucleon well inside the nucleus is almost constant due to geometry (Think of a "boondi" inside a "boondi ka laddoo"; practically each "boondi" is bound only to its neighbors. A nucleon which is at a distance more than the range of nuclear force has no contribution to the binding energy. If a nucleon can have p neighbors, its binding energy is proportional to p (say pk where k is some constant). Now if more nucleons are added in the volume, each such nucleon has p neighbors thus keeping binding energy per nucleon nearly constant. "Nearly" because nucleons on surface are surrounded by less neighbors and adding nucleons also increases the number of nucleons on surface. But number of nucleons on surface is

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much smaller than number inside the volume, so it does not alter the binding energy very significantly.

→ A very heavy nucleus (say $A=240$) has lower binding energy/nucleon. If it breaks in to two nuclei with $A=120$ each, it achieves higher binding energy per nucleon which is the basis of energy release in fission reactions.

→ Similarly if very light nuclei ($A \leq 10$) join to form heavy nuclei, the binding energy/nucleon for fused nucleus is greater. This forms the basis for release of energy through fusion reactions.

Nuclear Energy Levels:

Just as electrons are arranged in an atom in shells of varying energies, there is experimental evidence for arrangement of nucleons in various energy levels in the nucleon shells. The excitation of a nucleon from a lower level to a higher energy level makes a metastable/unstable nucleus which can go back to the ground state by emitting a gamma photon (very high energy photon).

Energy levels for the nuclide ^{28}Al, deduced from nuclear reaction experiments.

Nuclear Spin And Magnetism:

Many nuclides have intrinsic nuclear angular momentum or spin and an associated nuclear magnetic moment. Nuclear angular momenta are of the same order of magnitude as electron angular momenta but their magnetic moments are much smaller than typical atomic magnetic moments.

Nuclear Force:

There are two types of nuclear forces namely strong nuclear forces and weak nuclear forces. The strong force is attributed to exchange of "mesons" whose theory was developed by the Japanese Physicist, Hideki Yukawa. According to this theory the,

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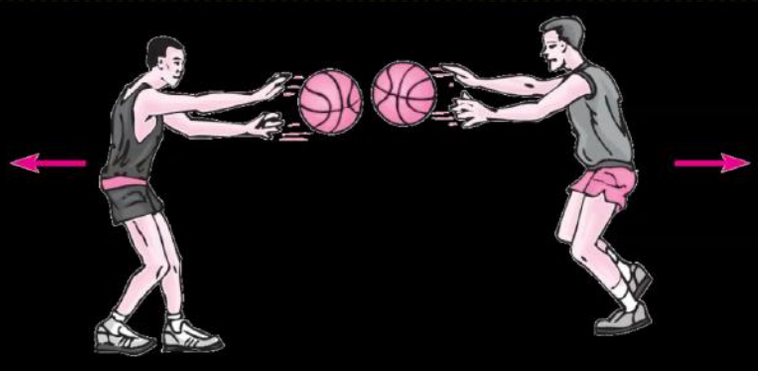
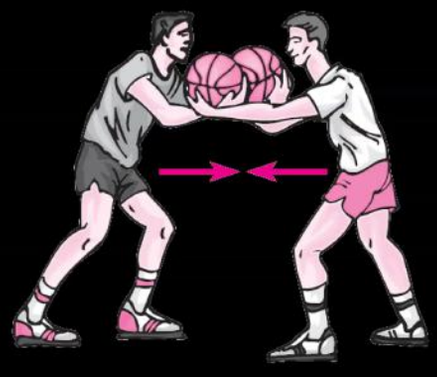
exchange of particles can create attractive as well as repulsive forces.

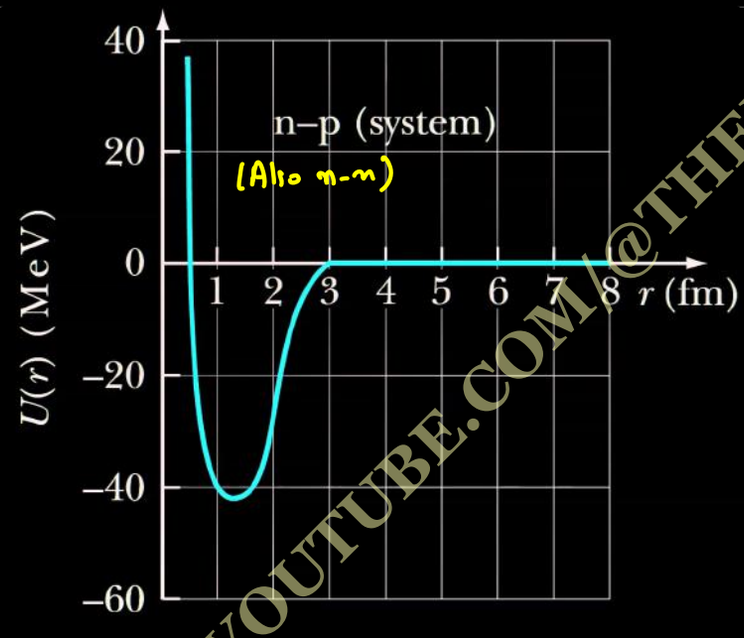


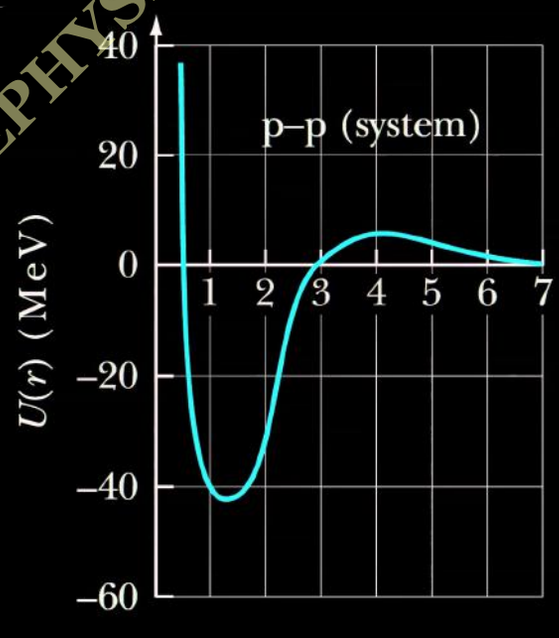
Repulsive force due to particle exchange



Attractive force due to particle exchange

The strong force is responsible for mutual attraction between the nucleons (also repulsion if the distance becomes too small). The weak nuclear force is mediated by Bosons and is involved in Beta Decay. The range of strong nuclear forces is about 2 fm.





The potential energy curves for internucleon separation are shown for interactions between protons (p-p) and protons & neutron (p-n). The two graphs look slightly different because there is a coulombic repulsion among protons while no coulombic repulsion between a neutron and another nucleon. The nuclear forces are much stronger than coulombic forces, which in turn are much stronger than gravitational forces

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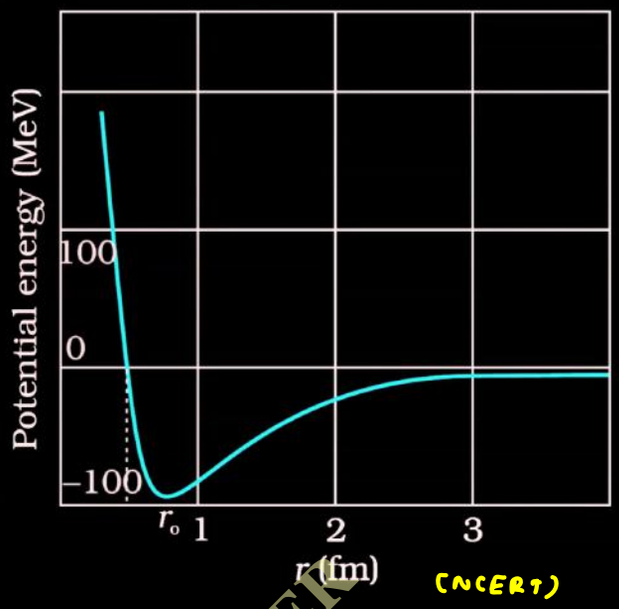
→ The nuclear force falls rapidly to zero as the distance between nucleons becomes more than a few femtometers. This leads to saturation of forces in medium and large sized nuclei leading to constancy of binding energy per nucleon.

→ The potential energy is minimum at a distance $r_0 = 0.8$ fm, at separation smaller than this, the force becomes repulsive while above this distance, the force is attractive.



(NCERT)

→ Nuclear force between n-n and n-p is same, it does not depend on electric charge of nucleons.

→ Unlike coulomb's law or Gravitation, there is no simple formula for nuclear forces

Binding Energy Of A Nucleus In Terms Of Mass Excess Terms:

The binding energy of a nucleus is given by the formula:

$$E_b = (Z m_H + (A-Z) m_n - M) c^2 \quad (1)$$

where m_H is mass of hydrogen atom, m_n is neutron mass and M is atomic mass of the atom corresponding to the given nucleus. Note that we are able to work with atomic masses because masses of electrons will cancel out. (Z hydrogen atoms will have same number of electrons as atom with atomic number Z). For purpose of calculations, a more convenient formula is:

$$E_b = (Z \Delta_H + (A-Z) \Delta_n - \Delta) c^2 \quad (2)$$

where Δ_H, Δ_n and Δ are mass excess of hydrogen atom, neutron and the atom corresponding to the given nucleus. Equation (2) directly follows from equation (1) as follows:

$\Delta_H = m_H - 1$ amu

$\Delta_n = m_n - 1$ amu $\quad \Delta = M - A$ amu

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Thus equation ① becomes, $E_b = \left(Z\,(\Delta_H + 1\,\text{amu}) + (A-Z)\,[\Delta_n + 1\,\text{amu}] - (\Delta + A\,\text{amu})\right)c^2$

Cancelling suitable terms, we get:

$$E_b = (Z\Delta_H + (A-Z)\Delta_n - \Delta)c^2 \quad \text{QED.}$$

EXAMPLE: IRODOV 6.255:

Find the binding energy of a nucleus consisting of equal numbers of protons and neutrons and having and radius one and a half times smaller than that of Al^{27} nucleus.

Solution: Here, we must first identify the nuclide and then use atomic mass chart. Now:

$R = r_0 A^{1/3}$ ① thus $\frac{2}{3}R = r_0 A'^{1/3}$ ②

Thus: $\left(\frac{A'}{A}\right)^{1/3} = \frac{2}{3} \Rightarrow \frac{A'}{A} = \frac{8}{27} \Rightarrow A' = 8$ This is ${}^{8}_{4}Be$

Now mass excess $\boxed{\Delta_H = 0.00783, \quad \Delta_n = 0.00867}$ $\Delta_{Be} = 0.00531$

Thus $E_b = (Z\Delta_H + (A-Z)\Delta_n - \Delta)c^2 = (4 \times 0.00783 + 4 \times 0.00867 - 0.00531)u \times c^2$

$= 0.0607\ uc^2$ now $1uc^2 = 931.5$ MeV, thus required E_b: 56.5 MeV

→ Energy released in a reaction is change in binding energy (Greater the final E_b more the energy released. This can be directly verified by writing partial reactions of splitting nuclei in to nucleons).

→ Binding Energy of isolated nucleons is zero (i.e for neutron/proton, $E_b = 0$)

EXAMPLE: IRODOV 6.256:

Making use of the tables of atomic masses, find:

(a) the mean binding energy per one nucleon in O^{16} nucleus;

(b) the binding energy of a neutron and an alpha-particle in a B^{11} nucleus;

(c) the energy required for separation of an O^{16} nucleus into four identical particles.

Given: $\Delta_{O^{16}} = -0.00509$, $\Delta_\alpha = 0.00260$ $\Delta_{B^{11}} = 0.00930$ $\Delta_{B^{10}} = 0.01294$

$\Delta_{Li^7} = 0.01601$

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Solution: (a) $\frac{E_b}{A}$ for O^{16} : $\frac{1}{16}\left[8 \times 0.00783 + 8 \times 0.00867 + 0.00509\right) uc^2$

$= \frac{0.13709 \times 931.5 \text{ MeV}}{16 \text{ nucleon}} = 7.98 \text{ MeV/nucleon.}$

(b) Here BE of neutron means energy required to remove a neutron. Which also means energy released upon combining a neutron with some nucleus to get the desired nucleus. Here:

$B^{11} \longrightarrow B^{10} + n$. Thus required binding energy is energy required in this reaction i.e $\left(\Delta_{B^{10}} + \Delta_n - \Delta_{B^{11}}\right) c^2$

Thus binding energy of neutron: $\left(0.01294 + 0.00867 - 0.00930\right) uc^2$

$= 11.46 \text{ MeV}$ Ans.

Similarly BE of α particle means energy required to pluck out an alpha particle or energy released upon adding α particle to a suitable nucleus to get desired nucleus. We have the reaction:

$^{11}B \longrightarrow {}^4_2He + {}^7_3Li$, thus required binding energy of α particle

$\left(\Delta_{He^4} + \Delta_{Li^7} - \Delta_{B^{11}}\right) uc^2 = \left(0.00260 + 0.01601 - 0.00930\right) \times 931.5$ MeV

$= 8.67$ MeV Ans.

(c) $^{16}O \longrightarrow 4\,{}^4_2He$, thus required energy:

$Q = \left(4\Delta_{He} - \Delta_{O^{16}}\right)_{uc^2} = \left[4 \times \left(0.00260\right) - \left(-0.00509\right)\right) 931.5 \text{ MeV}$

$= 14.42 \text{ MeV}$ Ans.

→ Note that since mass number is conserved, for finding the gained mass, we found the change in excess mass.

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EXAMPLE: IRODOV: 6.258:

Find the energy required for separation of a Ne^{20} nucleus into two alpha-particles and a C^{12} nucleus if it is known that the binding energies per one nucleon in Ne^{20}, He^{4}, and C^{12} nuclei are equal to 8.03, 7.07, and 7.68 MeV respectively.

Solution: Note that energy is released when binding energy increases and energy is required when binding energy decreases. Thus in any reaction, energy required is $E_{b_i} - E_{b_f}$ and energy released is $E_{b_f} - E_{b_i}$ (i: initial f: final). Our reaction is:

$$^{20}Ne \longrightarrow 2\,^{4}_{2}He + {}^{12}C .$$

Thus required energy: $20 \times e_{b_{Ne}} - 2e_{b_{He}} \times 4 - 12\, e_{b_{C^{12}}}$

$= 20 \times 8.03 - 2 \times 4 \times 7.07 - 12 \times 7.68 = 11.9$ MeV Ans.

EXAMPLE: IRODOV 6.259:

Calculate in atomic mass units the mass of

(a) a Li^{8} atom whose nucleus has the binding energy 41.3 MeV;

(b) a C^{10} nucleus whose binding energy per nucleon is equal to 6.04 MeV.

Solution: $E_b = (Z\Delta_H + (A-Z)\Delta_n - \Delta)uc^2$

$\Rightarrow \Delta = Z\Delta_H + (A-Z)\Delta_n - E_b/uc^2$

(a) $\Delta = 3 \times 0.00783 + 5 \times 0.00867 - 41.3/931.5 = 0.0225$

thus $M_{Li} = 8.0225\ u$

(b) $\Delta = 6 \times 0.00783 + 4 \times 0.00867 - \dfrac{6.04 \times 10}{931.5} = 0.01682$

$\Rightarrow m_{C^{10}} = 10.01682\ u$ Ans.

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EXAMPLE: IRODOV 6.260:

The nuclei involved in the nuclear reaction $A_1 + A_2 \rightarrow A_3 + A_4$ have the binding energies E_1, E_2, E_3, and E_4. Find the energy of this reaction.

Solution: Energy of a reaction means energy released in the reaction. Energy is released when final binding energy is more than initial binding energy. Thus it is difference $Q = E_{b\,product} - E_{b\,reactants}$

Thus released energy is simply $(E_3 + E_4) - (E_1 + E_2)$ Ans.

EXAMPLE: IRODOV 6.264:

Find the energy of the reaction $Li^7 + p \rightarrow 2He^4$ if the binding energies per nucleon in Li^7 and He^4 nuclei are known to be equal to 5.60 and 7.06 MeV respectively.

Solution: $Q = E_{b\,products} - E_{b\,reactants}$, also isolated nucleon have zero binding energies. Thus, $Q = 2 \times 7.06 \times 4 - 7 \times 5.6$ MeV $= 17.28$ MeV

EXAMPLE: IRODOV 6.267:

Making use of the tables of atomic masses, find the velocity with which the products of the reaction $B^{10}(n, \alpha)\,Li^7$ come apart; the reaction proceeds via interaction of very slow neutrons with stationary boron nuclei. $\Delta_{Li^7} = 0.0601$, $\Delta_{B^{10}} = 0.01294$ $\Delta_\alpha = 0.00260$ $\Delta m = 0.00867$

Solution: Here $B^{10}(n,\alpha)\,Li^7$ means the reaction: ${}^{10}B + n \rightarrow {}^{7}Li + \alpha$

Now since initial momentum is $\neq 0$ $\therefore$ final momentum must also be zero. This means, released energy will be distributed in inverse ratio of masses. ($\because$ p is same Δ $KE = p^2/2m$). Now Energy released:

$$(\Delta_{reactants} - \Delta_{products})\,uc^2 = (0.01294 + 0.00867 - 0.0601 - 0.00260) \times 931.5\ \text{MeV}$$

$$= 2.79\ \text{MeV} = \frac{1}{2}\mu v_{rel}^2 = \frac{1}{2} \times \left(\frac{7 \times 4}{7+4}\right) \times 931.5\ \frac{\text{MeV}}{c^2} \times v_{rel}^2$$

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$$\Rightarrow \quad v_{rel} = c\sqrt{\frac{2 \times 2.79 \times 11}{931.5 \times 28}} = 1.46 \times 10^7 \ ms^{-1}$$

Thus $v_{Li} = \left(\frac{1/7}{1/7+1/4}\right) \times 1.46 \times 10^7 \ ms^{-1} = 5.31 \times 10^6 \ ms^{-1}$ Ans.

$$v_{\alpha} = \frac{7}{11} \times 1.46 \times 10^7 = 9.29 \times 10^6 \ ms^{-1} \quad \text{Ans.}$$

RADIOACTIVE DECAY LAW: As discussed earlier, elements heavier than a certain mass number spontaneously decay in to other species. The decay process is random i.e it is hard to predict which nucleus will decay next, however, the numbers decaying per second is directly proportional to the absolute number of nuclei. Thus:

$\frac{dN}{dt} \propto N$ when N is the absolute number of active Nuclei.

Thus we can write $\boxed{\frac{dN}{dt} = -\lambda N}$ (1), Here λ is called disintegration constant and is a property of the given nuclide.

→ λ is also known as decay constant or disintegration constant.

ACTIVE NUCLEI AT A GENERAL TIME: We can integrate the equation (1) above to get active nuclei at a general time. Let N_0 be the number of active nuclei at $t = 0$ then:

$$\frac{dN}{dt} = -\lambda N \quad \Rightarrow \quad \frac{dN}{N} = -\lambda dt, \text{ now integrating:}$$

$$\int_{N_0}^{N} \frac{dN}{N} = \int_0^t -\lambda dt \quad \Rightarrow \quad \ln \frac{N}{N_0} = -\lambda t \quad \Rightarrow \quad \boxed{N = N_0 e^{-\lambda t}} \ (2)$$

ACTIVITY OF A SAMPLE: In many radioactive decays, often a particle is emitted (such as α particle, β particle or a γ photon). The number of particles registered by a detector per unit time is called activity of the sample. If the products of a decay are not radioactive then the activity is simply the rate of decay of the sample i.e $A = |dN/dt|$.

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Thus using equation ① and ②, at $t=0$, initial activity:

$A_0 = \lambda N_0$ and at a general time, the activity:

$A = \lambda N_0 e^{-\lambda t} \Rightarrow$ $\boxed{A = A_0 e^{-\lambda t}}$

→ Activity of any sample is measured in Becquerels (Bq), 1 Becquerel is simply 1 decay/sec.

→ Another commonly used unit of activity is Curie (Ci). $\boxed{1 \text{ Curie} = 3.7 \times 10^{10} \text{ Bq}}$ 1 Ci is the activity of 1g of naturally occuring Radium.

HALF LIFE OF A SAMPLE: Time taken by a sample for reduction of its active nuclei to half of its initial value is called half life. Thus by definition if N_0 is the initial number of active nuclei then after half life (T_H) the active nuclei number becomes $N_0/2$. Thus:

$$\frac{N_0}{2} = N_0 e^{-\lambda T_H} \Rightarrow \ln 2 = \lambda T_H$$

$$\Rightarrow \boxed{T_H = \frac{\ln 2}{\lambda}} \qquad \boxed{\ln(2) = 0.693}$$

MEAN LIFE: The time after which the number of active nuclei reduces to $1/e$ times its initial value is called mean life. This is also the statistical average of lifetime of all the nuclei in the species. The two definitions of mean life (τ) can be shown to be equivalent. From first definition:

$$\frac{N_0}{e} = N_0 e^{-\lambda \tau} \Rightarrow 1 = \lambda \tau \Rightarrow \boxed{\tau = \frac{1}{\lambda}} \quad ①$$

Now imagine a small time window from t to $t+dt$. The number of nuclei dying in this time window: $|dN| = \lambda N dt$

Thus sum of the ages of these nuclei at the time of their decay is $\lambda N t\, dt$. (These were alive for a total time t).

Thus mean life: $\tau = \frac{1}{N_0} \int_0^\infty \lambda N t\, dt$

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$$= \frac{1}{N_0}\int_0^\infty \lambda N_0 e^{-\lambda t} t\, dt = \lambda \int_0^\infty t e^{-\lambda t} dt$$, Now integrating by parts: this is: $\lambda\left[t\frac{e^{-\lambda t}}{-\lambda} - \int \frac{e^{-\lambda t}}{-\lambda} dt\right]_0^\infty$

$= \lambda\left[\frac{t e^{-\lambda t}}{-\lambda} - \frac{e^{-\lambda t}}{\lambda^2}\right]_0^\infty$, Now as $t \to \infty$, $te^{-\lambda t} \to 0$

$= \frac{1}{\lambda}$, thus $\tau = \frac{1}{\lambda}$, this is same result as obtained in equation ①, QED.

→ Thus, $T_H = \tau \ln 2$ and $N = N_0 e^{-t/\tau}$

PROBABILITY OF SURVIVAL AND DECAY: The decay process is completely random so we can never be 100% sure about any particular nucleus as to whether or not it will decay, however we can say that probability of survival is equal to the fraction surviving at a general time thus probability of survival: $\eta_s = \frac{N}{N_0} = e^{-\lambda t}$ and probability of decay:

$$\boxed{\eta_d = 1 - e^{-\lambda t}}$$

EXAMPLE: Suppose we have a large sample of radioactive nuclei. Out of that sample, we pick four nuclei. What is the probability that exactly two nuclei would have decayed out of these in one half life?

Solution: Let us number the nuclei 1 to 4. We want exactly two nuclei decayed, e.g. We want 1&2 decayed which also means 3 & 4 should be undecayed. Since in 1 half life, probability of survival and decay are each equal to 1/2 thus probability of 1&2 decayed is $\underset{①}{\frac{1}{2}} \times \underset{②}{\frac{1}{2}} \times \underset{③}{\frac{1}{2}} \times \underset{④}{\frac{1}{2}}$ i.e 1/16.

Now number of ways of choosing any two nuclei out of 4 is $^4C_2 = 6$ thus required probability is $\frac{1}{16} \times 6 = 3/8$ Ans.

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Example: Irodov 6.216:

How many beta-particles are emitted during one hour by 1.0 μg of Na^{24} radionuclide whose half-life is 15 hours? (Each decay emits 1 β particle)

Solution:

Indirectly the question is asking how many nuclei decayed in 1 hour. Now $N = N_0 e^{-\lambda t}$ thus decayed nuclei: $N_0 - N$

i.e $N_0 (1 - e^{-\lambda t}) = N_0 (1 - e^{-t/\tau})$, $T_H = \tau \ln 2$

thus: $N_\beta = N_0 (1 - e^{-(t/T_H)\ln 2}) = N_0 (1 - 2^{-t/T_H}) = \frac{1\,\mu g}{24\ g/mol} (1 - 2^{-1/15})$

$= 1.13 \times 10^{15}$ Ans.

$$\boxed{N = N_0\, 2^{-t/T_H}}$$

Example: Irodov 6.217:

To investigate the beta-decay of Mg^{23} radionuclide, a counter was activated at the moment $t = 0$. It registered N_1 beta-particles by a moment $t_1 = 2.0$ s, and by a moment $t_2 = 3t_1$ the number of registered beta-particles was 2.66 times greater. Find the mean lifetime of the given nuclei. (Each Decay Emits 1 Beta Particle)

Solution: $N_1 = N_0 (1 - e^{-t_1/\tau})$, $N_2 = N_0 (1 - e^{-t_2/\tau})$

but $t_2 = 3t_1$ thus, $N_2 = N_0 (1 - e^{-3t_1/\tau})$ thus:

$$\frac{N_2}{N_1} = 2.66 \approx \frac{8}{3} = \left(\frac{1 - e^{-3t_1/\tau}}{1 - e^{-t_1/\tau}}\right) = \frac{1 - e^{-6/\tau}}{1 - e^{-2/\tau}}, \text{ but } e^{-2/\tau} = x$$

$$\Rightarrow \frac{8}{3} = \frac{1 - x^3}{1 - x} = (1 + x^2 + x) = \frac{8}{3}, \text{ Solving we get } x = 0.884$$

$$\Rightarrow \frac{-2}{\tau} = \ln 0.884 \Rightarrow \tau = \frac{-2}{\ln 0.884} = 16.28 \text{ seconds}$$

Types Of Radioactive Decay:

Alpha Decay: In this type of decay, an unstable nucleus emits an alpha particle thus its Z decreases by 2 and A decreases by 4:

$$^{A}_{Z}X \longrightarrow {}^{A-4}_{Z-2}Y + {}^{4}_{2}He$$

The nucleus before decay is called parent nucleus and that

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after decay is called daughter nucleus. For example:

$$^{212}_{83}Bi \longrightarrow {}^{208}_{81}Tl + {}^{4}_{2}He \quad \text{or}$$

$$^{238}_{92}U \longrightarrow {}^{234}_{90}Th + {}^{4}_{2}He \quad \text{(NCERT)}$$

Alpha decay occurs in all nuclei with $A > 210$

Q-VALUE: The difference between rest mass energy of initial constituents and final products is called Q value of any nuclear reaction. Thus, $Q = U_i - U_f$ where U_i and U_f are initial and final rest mass energies. For Alpha decay, the Q value is given as:

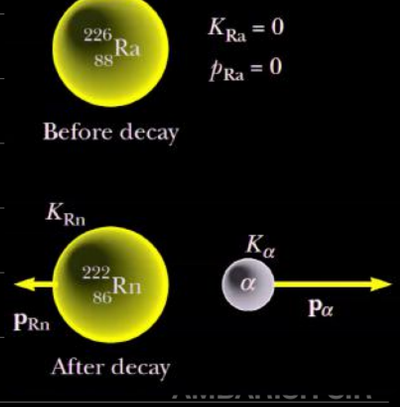
$$Q = (M_x - M_y - M_\alpha)c^2$$ where M_x, M_y and M_α are atomic masses of nuclei $^{A}_{Z}X$, $^{A-4}_{Z-2}Y$ and $^{4}_{2}He$.

→ Here instead of Atomic masses we could use same formula with nuclear masses as well since number of electrons in element X will be sum of number of electrons in Y and $^{4}_{2}He$. (Since Z is conserved).

→ Q values are always defined in terms of loss of nuclear masses (Atomic masses are used just for sake of convenience).

ENERGY SHARE: The energy released in α decay is shared in the inverse ratio of masses between α particle and daughter nucleus. Thus if Q is the energy released then $E_\alpha = \frac{M_y}{M_y + M_\alpha} Q$ and $E_y = \frac{M_\alpha}{M_y + M_\alpha} Q$.

The momenta of alpha particle is equal to that of the daughter. The total released energy in the decay is equal to $\frac{1}{2}\mu v_{rel}^2$ where μ is the reduced mass of α particle and the daughter.



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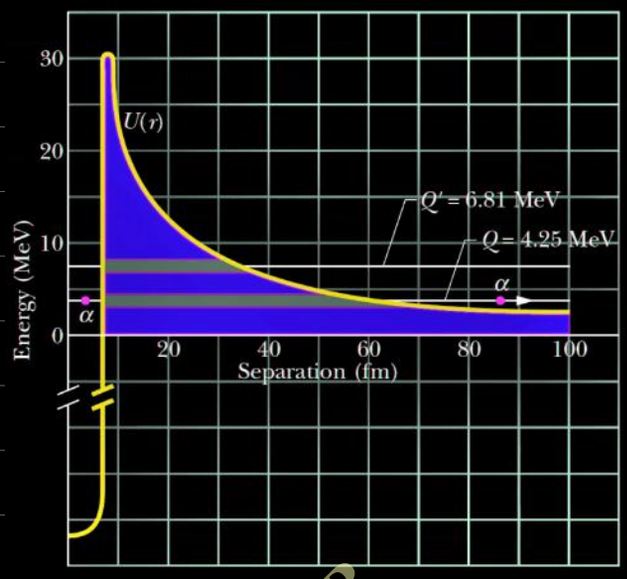
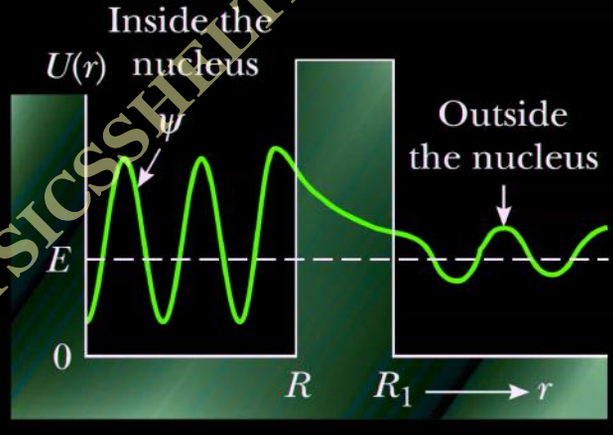
Barrier Tunneling: When a nucleus emits an alpha particle it has to overcome the attraction of the daughter nucleus before the coulombic repulsion takes over. The yellow curve represents the potential energy as a function of separation between α particle and the daughter nucleus for U^{232} and U^{238} (The graph is same due to same nuclear charge). In α decay of U^{238}, $Q = 4.25$ MeV while that for U^{232}, $Q = 6.81$ MeV, If we look at the potential Energy Graph, the α particle has to overcome a potential barrier of 30 MeV. Thus classically α decay is not possible. (The Grey regions in the graph are classically forbidden for the alpha particle), However according to quantum mechanics the problem is modeled as particle in a finite potential well and has a finite wavefunction outside the well. Thus α decay proceeds through barrier tunneling. Infact presence of the energy barrier is the reason for very long half life of U^{238} nuclide. The alpha particle can be assumed rattling back and forth within the nucleus due to energy barrier. It must arrive at the barrier 10^{38} times before it succeeds in tunneling through the barrier. This is about 10^{21} per second for 4×10^9 years (i.e the age of earth) and we are just waiting outside to count the alpha particle which manage to escape. (Half life of U^{238} is 4.5×10^9 years).





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Beta Decay: This is a decay process in which either a neutron is converted in to a proton or a proton is converted in to a neutron to improve the N/Z ratio (to achieve stability). The beta decay proceeds through weak nuclear forces (involves intermediate particles like W and Z Bosons). When a neutron is converted to proton, an electron and a new particle anti-neutrino ($\bar{\nu}$)

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are created and emitted from the nucleus. Thus:

$$n \longrightarrow p + e + \bar{\nu}.$$

The antineutrino is supposed to have a zero rest mass (like photons) and a spin $1/2$. The emitted electron is called a β particle and a stream of such electrons is called β rays.

The β^- decay process may be represented as:

$${}^{A}_{Z}X \longrightarrow {}^{A}_{Z+1}Y + e + \bar{\nu} \qquad \text{e.g.} \quad {}^{32}_{15}P \longrightarrow {}^{32}_{16}S + e^- + \bar{\nu} \quad \text{(NCERT)}$$

Here $U_i = (M_x - Z m_e)c^2$ and $U_f = [M_y - (Z+1)m_e + m_e]c^2$

Thus $\boxed{Q_{\beta^-} = U_i - U_f = (M_x - M_y)c^2}$. Thus for β^- decay, we can simply work with atomic masses of X and Y and ignore the mass of electron.

BETA PLUS DECAY: If an unstable nucleus has excess protons than required for stability, a proton converts itself to a neutron, positron and a neutrino:

$$p \longrightarrow n + e^+ + \nu$$

The positrons has same mass as an electron but a positive charge. It is called an anti-particle of electron. (Similarly ν and $\bar{\nu}$ are anti-particles). The β^+ decay process is represented as:

$${}^{A}_{Z}X \longrightarrow {}^{A}_{Z-1}Y + e^+ + \nu \qquad \text{for example} \quad {}^{22}_{11}Na \longrightarrow {}^{22}_{10}Ne + e^+ + \nu$$

Now let us find Q value for such a reaction:

Here $U_i = (M_x - Z m_e)c^2 \qquad U_f = [M_y - (Z-1)m_e + m_e]c^2$

Thus $\boxed{Q = U_i - U_f = (M_x - M_y - 2m_e)c^2}$ Notice that here we need to subtract two electron masses and can't simply subtract the atomic masses unlike α and β^- decays.

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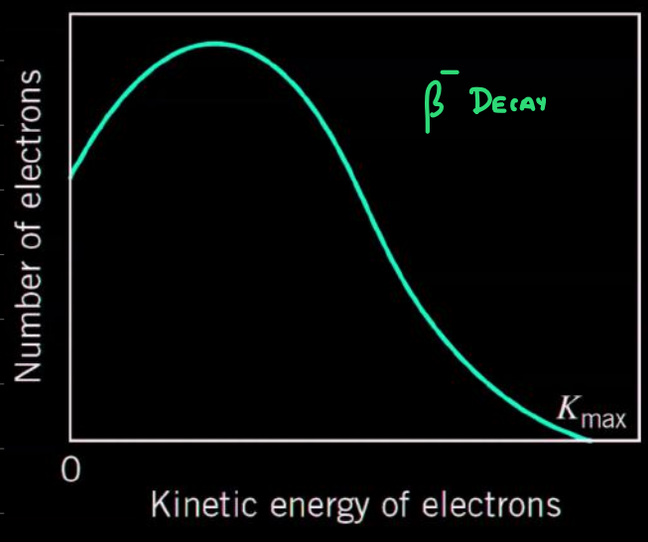
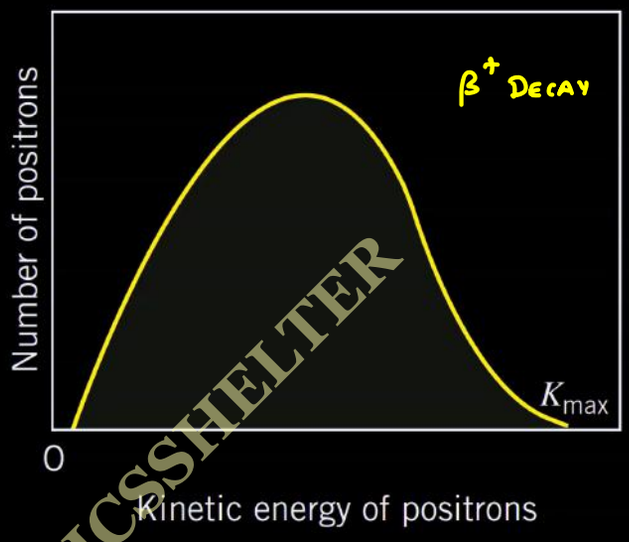
→ A free neutron spontaneously decays to a proton (T_H = 10.2 minutes) while free protons are stable (β^+ decay happens only in the nucleus).

Energy Spectrum In β^- And β^+ Decays:





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At one point of time, the existence of neutrinos was not known. When the energies of beta particles in the beta decay were studied it was found that unlike alpha decay, the energies of beta particles varied significantly. This was a puzzle for the physicists. The existence of an energy spectrum meant that there must be a third particle sharing the energy in the reaction apart from the beta particle and the daughter nucleus. Thus Pauli first suggested the existence of neutrinos (The name was given by Enrico Fermi. The name neutrino roughly means the neutral little one). The energy spectra for β^- and β^+ particles are presented in the graph above. In these spectra it is remarkable that:

→ The β^- energy spectrum starts from zero energy i.e there are some finite number of particles coming out at nearly zero energies. This is because β^- particles have to overcome nuclear coulombic attraction as they move away from the daughter nucleus. In this process they gain electrostatic energy and hence lose kinetic energy.

→ β^+ energy spectrum starts from non-zero energy. This is because they experience a repulsion from nucleus and hence lose electrostatic potential energy and thus gain kinetic energy.

→ Neutrinos can have a minimum of zero energy to a maximum energy equal to Q value of the reaction.

→ The Q value of reaction is equal to K_{max} (This is because electron is much lighter than the Daughter nucleus and when the neutrino does not share any energy, the energy is shared in inverse mass ratios between the daughter and beta particle thus practically in that case entire energy will be carried away by the beta particle).

→ Neutrinos are very difficult to detect because they almost do not interact with matter and pass through it unaffected. At every moment countless number of neutrinos pass through earth without producing any noticeable effect. Since neutrinos do not interact with matter, the energy carried by neutrinos cannot be put to any use, it is essentially stolen away from us. In laboratory the neutrinos always take away a significant amount of energy and electron almost always has an energy less than K_{max}.

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K- SHELL ELECTRON CAPTURE: In nucleon species with excess protons, a process which competes with β^+ decay is K shell electron capture. In this process, the nucleus captures a K shell electron and converts a proton in to a neutron. Thus: $p + e = n + \nu$. The process can be written as:

$${}^{A}_{Z}X + e \longrightarrow {}^{A}_{Z-1}Y + \nu$$

Here the Q value can be found as follows: $U_i = (M_x - Z m_e + m_e)c^2$ $U_f = [M_y - (Z-1)m_e]c^2$

thus Q value:

$$Q = U_i - U_f = [M_x - M_y]c^2 \quad : \quad \boxed{Q = [M_x - M_y]c^2}$$

→ Here again electron masses do not feature in Q value and we can just work with atomic masses of X and Y.

→ K-electron capture is followed by emission of X-Rays due to the vacancy created in the K-Shell.

→ Practically entire energy in K capture is carried away by the neutrino because of large mass of the daughter nucleus.

EXAMPLE: Knowing the mass of daughter atom and β^- decay energy Q find the atomic mass of (a) ^6He which undergoes a β^- decay with Q value 3.5 MeV (b) ^{22}Na undergoing β^+ decay with Q value 1.83 MeV

Given $m(^6_3Li) = 6.01512u$, $m(^{22}_{10}Ne) = 21.99144u$, $m_e = 0.000548 6u$

Solution: (a) $^6He \longrightarrow {}^6_3Li + e^- + \bar{\nu}$, $Q = (M_{He} - M_{Li})c^2$

$$\Rightarrow M_{He} = \frac{Q}{c^2} + M_{^6_3Li} = \frac{3.5\ MeV}{c^2} \times \frac{1u\,c^2}{931.5\ MeV} + 6.0152u = 6.0189u$$ Ans.

(b) $^{22}Na \longrightarrow {}^{22}Ne + e^+ + \nu$ $\quad Q = [M_{Na} - M_{Ne} - 2m_e]\times c^2$

$$\Rightarrow M_{Na} = \frac{Q}{c^2} + M_{Ne} + 2m_e = \frac{1.83\ MeV}{c^2} \times \frac{1uc^2}{931.5\ MeV} + 21.99144 + 2\times(0.000586)$$

$= 21.9946u$ Ans

EXAMPLE: ^{37}Ar undergoes K shell capture after which daughter nuclei are formed in ground state. Neglecting the binding energy of the K electron determine the KE and velocity of daughter nucleus.

Given $m(^{37}_{17}Cl) = 36.9659u$, $m(^{37}Ar) = 36.9668u$ (neutrino can be treated like photon).

Solution:

$$^{37}Ar + e \longrightarrow {}^{37}_{17}Cl + \nu$$

$Q = (M_{Ar} - M_{Cl})\times c^2 = 0.0009 \times 931.5\ MeV = 0.838\ MeV$

Now practically entire energy is taken by neutrino thus neutrino momentum: $p_\nu = \frac{Q}{c} = p_{Cl}$ Now KE of Cl: $K = \frac{p^2}{2M_{Cl}} = \frac{Q^2}{2Mc^2}$

$$\Rightarrow KE_{Cl} \simeq \frac{(0.838\ MeV)^2}{2\times 37\times 931.5\ MeV} \simeq 10eV$$ Ans.

Now speed $v = \frac{p}{M} = \frac{Q}{Mc} = \frac{0.838\ MeV}{37\times 931.5\ MeV/c^2 \times c} = 2.43\times10^{-5}c \simeq 7\times10^3 ms^{-1}$ Ans.

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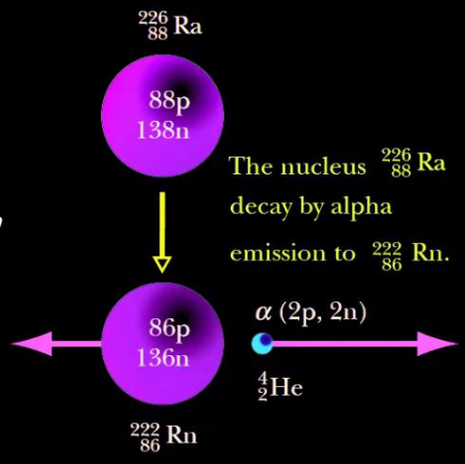
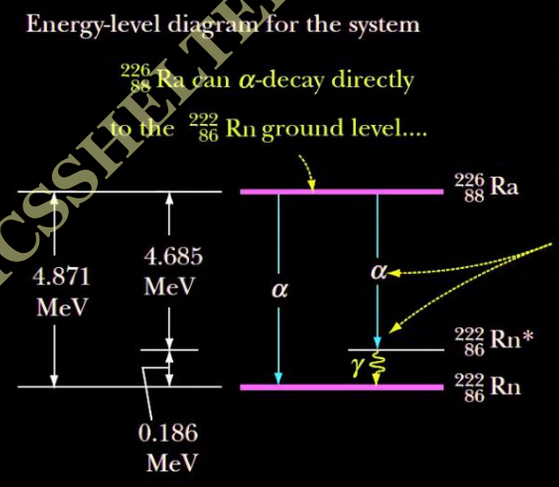
GAMMA DECAY: As we have seen earlier that energy levels in a nucleus are quantized thus similar to excited states of atoms there can exist excited states of a nucleus according to laws of quantum mechanics. In excited states a nucleus can get de-excited to ground state by emitting one or more high energy photons called Gamma rays. Typical energy of Gamma photons is from 10keV to 5MeV (Overlaps with x-rays). The process is known as a Gamma decay. For example alpha particles emitted from Ra^{226} have two possible kinetic energies (including that of daughter nucleus) i.e either 4.784 or 4.602 MeV. When alpha particle of smaller energy is emitted, Rn^{222} nucleus is left in an excited state, which releases rest of the energy required to reach ground state (i.e Gamma photon of energy 0.186 MeV). (See the energy diagram above).





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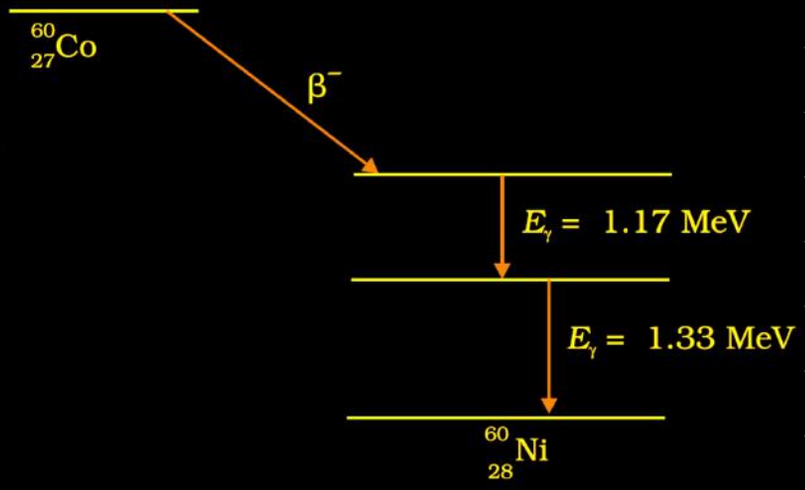
→ Usually in Gamma Decays practically entire kinetic energy is taken away by the Gamma photons.

→ Another example of Gamma decay mentioned in NCERT is that of Co^{60} to Ni^{60} which passes through two unstable states of the Nickel nucleus. Thus there are two successive emissions of Gamma photons (See adjacent Energy diagram).



Example: Irodov 6.236:

The alpha-decay of $P0^{210}$ nuclei (in the ground state) is accompanied by emission of two groups of alpha-particles with kinetic energies 5.30 and 4.50 MeV. Following the emission of these particles the daughter nuclei are found in the ground and excited states. Find the energy of gamma-quanta emitted by the excited nuclei.

Solution: In alpha decay, energy released is distributed in inverse ratio of masses thus $E_\alpha = Q\left(\frac{m_D}{m_D + m_\alpha}\right)$ (D: Daughter).

Thus $Q = \left(1 + \frac{m_\alpha}{m_D}\right) E_\alpha$, So for first reaction, $Q_1 = \left(1 + \frac{4}{206}\right) \times 5.3$

similarly for second reaction, $Q_2 = \left(1 + \frac{4}{206}\right) \times 4.5$

The energy of Gamma photon must be the difference of these two Q values, thus $E_\gamma = \left(1 + \frac{4}{206}\right)(5.3 - 4.5) \approx 0.82$ MeV Ans.

Example: Irodov 6.244:

Passing down to the ground state, excited Ag^{109} nuclei emit either gamma quanta with energy 87 keV or K conversion electrons whose binding energy is 26 keV. Find the velocity of these electrons.

Solution: Here Gamma photon energy is being consumed in knocking out K shell electrons. Thus KE of electrons must be the difference in Gamma photon energy and binding energy of K shell electron. Thus KE of electrons: $KE = 87\,keV - 26\,keV = 61\,keV$. The rest mass energy of an electron is 511 keV and KE is about 10% of the rest mass energy so we will get approximately correct results even without using relativistic formulae. Though we will do both types of calculations.

Non-relativistic: $KE = 61\,keV = \frac{1}{2}mv^2 = \frac{1}{2} \times 511\,\frac{keV}{c^2} \times v^2$

$$\Rightarrow v = c\sqrt{\frac{122}{511}} \approx 1.47 \times 10^8\ m/s.$$

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Relativistic: $E = m_0c^2 + KE = 511\,keV + 61\,keV = mc^2 = 572\,keV$

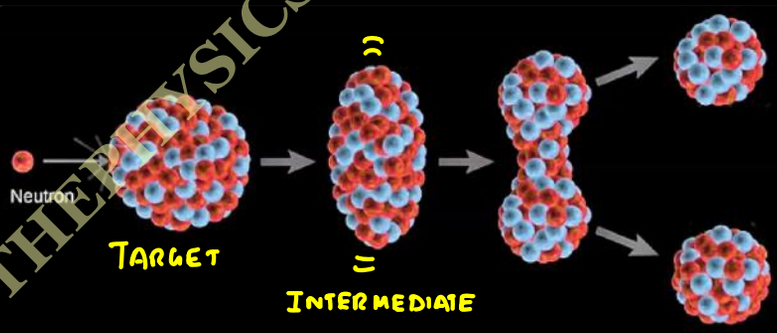
Thus $\frac{m_0}{\sqrt{1 - v^2/c^2}} \times c^2 = 572\,keV \Rightarrow 1 - \frac{v^2}{c^2} = \left(\frac{511}{572}\right)^2$ ($\because m_0c^2 = 511\,keV$)

$\Rightarrow v = c\sqrt{1 - \left(\frac{511}{572}\right)^2} = 1.35 \times 10^8\ ms^{-1}$

→ Inadov Answer is almost matching with non-relativistic calculation.

→ If KE is comparable to rest mass energy, Relativistic calculations are generally used however if $KE \ll TE$, non-relativistic calculations do a fairly good job. In our case we get an error of about 9%.

Threshold Energy Of Endoergic Reactions:



The reactions in which final mass is more than the initial mass are called endo-ergic reactions. For such reactions, Q value is negative. The energy for such a reaction is typically provided by bombarding a target nucleus (mass M) by a projectile (mass m). Here due to conservation of momentum, the products formed are not at rest in the lab frame, thus the projectile must have an energy over and above |Q| to account for KE of the particles in the lab frame. The minimum required energy in the lab frame to start such a reaction is called threshold energy (E_{th}). We will derive an expression for it using concept of CM frame. Recall that CM frame is a zero momentum frame, thus in CM frame the required energy will be exactly equal to |Q| in the limiting case. Thus:

$E_{cm} = \frac{1}{2}\mu v_{rel}^2 = |Q|$ (1) also from lab frame,

$E_{th} = \frac{1}{2} m v_{rel}^2$ (2) From (1) & (2):

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$$\frac{E_R}{|Q|} = \frac{m}{\mu} = \frac{m(M+m)}{Mm} = \left(1+\frac{m}{M}\right) \quad \text{thus,}$$

$$\boxed{E_{th} = |Q|\left[1+\frac{m}{M}\right] = -Q\left[1+\frac{m}{M}\right]}$$ ← Threshold Energy for Endoergic Reactions.

EXAMPLE: Calculate the threshold Kinetic Energy for the reaction $p + {}^3H \longrightarrow {}^2H + {}^2H$ if (a) Protons are incident on ${}^3H$ at rest (b) Tritons are incident on protons at rest. Given $\Delta^1H = 0.007825$ $\Delta^3H = 0.016049$ $\Delta^2H = 0.014102$.

Solution: $Q = \Delta p + \Delta^3H - 2\Delta^2H = (0.007825 + 0.16049 - 2\times(0.014102)) \times 931.5\ MeV$

$= -4.033\ MeV$

(a) $E_{th} = \left[1+\frac{m_p}{m_t}\right] \times 4.033 = \left(1+\frac{1}{3}\right) \times 4.033 = 5.38\ MeV$ Ans

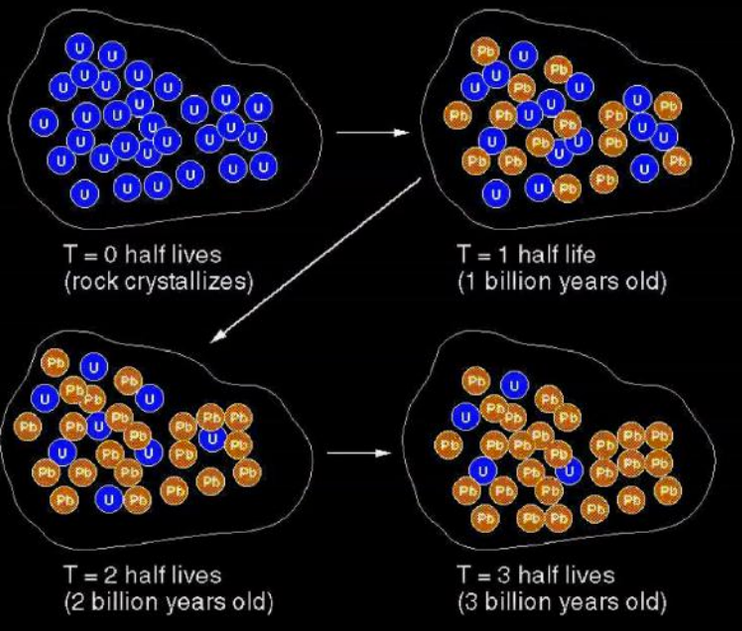
(b) $E_R = \left(1+\frac{m_t}{m_p}\right) \times 4.033 = (1+3) \times 4.033 = 16.10\ MeV$ Ans.

t: triton.

→ Lesser energy is required if a lighter particle in incident on a heavy nucleus than the other way round.

RADIOACTIVE DATING:

Heavy elements such as Uranium are formed during Supernova explosion. After the explosion, Uranium U^{235} forms crystals which might be present in various rocks. U^{235} gradually decays to Pb^{207} with a half life of about 700 million years. Thus knowing the ratio of Pb to U in the rocks one can estimate the age of the rock at the time of its formation (This assumes



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that at the time of formation, it was completely U^{235}). Similarly, K^{40} decays to Ar^{40}, a stable isotope, thus knowing the ratio of K^{40} to Ar^{40} also one can date a rock. (Half life is 1.25×10^9 years).

RADIO-CARBON DATING: The radio-nuclide C^{14} is produced at a constant rate in upper atmosphere due to bombardment of atmospheric nitrogen by cosmic rays. Thus atmosphere maintains a constant ratio of C^{12} to C^{14} (There is about 1 C^{14} atom for every 10^{13} atoms of ordinary C^{12} atoms). Since living animals and plants derive carbon in their bodies directly or indirectly from the atmosphere (through photosynthesis / vegetable / carnivorous diets). their bodies also have same C^{14} : C^{12} ratio when they are alive. Once the person or living being dies, C^{14} begins to decay to N^{14} (Beta Decay) with a half life of 5700 ± 30 years. Thus knowing the ratio of active C^{14} nuclei to C^{12} nuclei in a fossil sample or derivatives of the dead bodies of organisms (wood / skin / cereals / bones / hair etc) one can estimate the age of the sample i.e approximate time when the organism was alive (exchanging carbon with surrounding).

EXAMPLE: IRODOV : 6.222:

Determine the age of ancient wooden items if it is known that the specific activity of C^{14} nuclide in them amounts to 3/5 of that in lately felled trees. The half-life of C^{14} nuclei is 5570 years.

Solution: $A = A_0 e^{-t/\tau} = A_0 e^{-t \ln 2 / T_H}$

$\Rightarrow A = A_0 2^{-t/T_H} \Rightarrow \ln \frac{3}{5} = \frac{-t}{T_H} \ln 2 \Rightarrow t = T_H \ln(5/3) / \ln(2)$

$= 5570 \times (\ln 5 - \ln 3) / \ln(2) = 4105$ Years Ans.

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Example: Irodov: 6.225:

A small amount of solution containing Na^{24} radionuclide with activity $A = 2.0 \times 10^3$ disintegrations per second was injected in the bloodstream of a man. The activity of $1\,cm^3$ of blood sample taken $t = 5.0$ hours later turned out to be $A' = 16$ disintegrations per minute per cm^3. The half-life of the radionuclide is $T = 15$ hours. Find the volume of the man's blood.

Solution: Here since the radioactive sample is mixed in blood stream, it get diluted, so the immediate activity of blood per unit volume will be reduced depending on volume of blood in person's body. This activity will keep reducing with time due to radio decay.
Now let blood volume be $V\,cm^3$ then activity $/cm^3$ after t time must be:

$$A' = \frac{A}{V} e^{-t\ln 2/T_H} \Rightarrow V = \frac{A}{A'} 2^{-t/T_H} = \frac{2\times10^3}{(16/60)\,/cm^3} \times 2^{-1/3}$$

$= 5.95 \times 10^3\ cm^3 \approx 6$ litre Ans.

Radioactive Series Reactions: Successive Growth & Decay Process

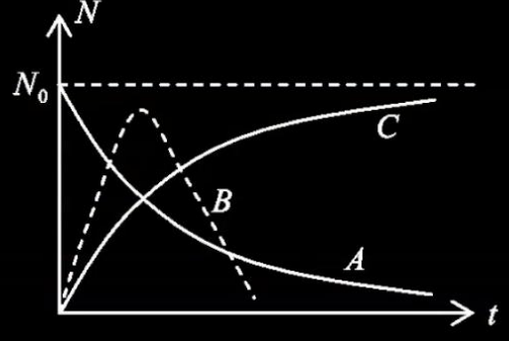
In a successive growth/decay process $A \to B \to C$, element C is a stable nucleus. The following parameters are given for the process:

$$A \xrightarrow{\lambda_1} B \xrightarrow{\lambda_2} C$$

$t = 0$	N_0	0	0
$t = t$	N_1	N_2	N_3

where λ_1 and λ_2 are decay constant, N_0 is the concentration of A at $t = 0$ and N_1, N_2, N_3 are concentration of A, B, C at any time t.

Thus $\boxed{N_1 = N_0 e^{-\lambda_1 t}}$ (1)



Rate equation for B: $\dfrac{dN_2}{dt} = \lambda_1 N_1 - \lambda_2 N_2$

$$\Rightarrow \frac{dN_2}{dt} + \lambda_2 N_2 = \lambda_1 N_0 e^{-\lambda_1 t}$$

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Multiply both side by $e^{\lambda_2 t}dt$ and then integrate $\Rightarrow N_2 e^{\lambda_2 t} = \frac{\lambda_1 N_0 e^{(\lambda_2-\lambda_1)}}{\lambda_2-\lambda_1} + K$

At $t=0, N_2=0 \Rightarrow K = -\frac{\lambda_1 N_0}{\lambda_2-\lambda_1}$, thus $\boxed{N_2 = \frac{\lambda_1 N_0}{\lambda_2-\lambda_1}\left[e^{-\lambda_1 t} - e^{-\lambda_2 t}\right]}$ (2)

If C is a stable nucleus, the rate of decay of atoms of B into C i.e $\frac{dN_3}{dt}$ is given by,

$$\frac{dN_3}{dt} = -\lambda_2 N_2 \qquad (\text{At } t=0,\ N_3=0)$$

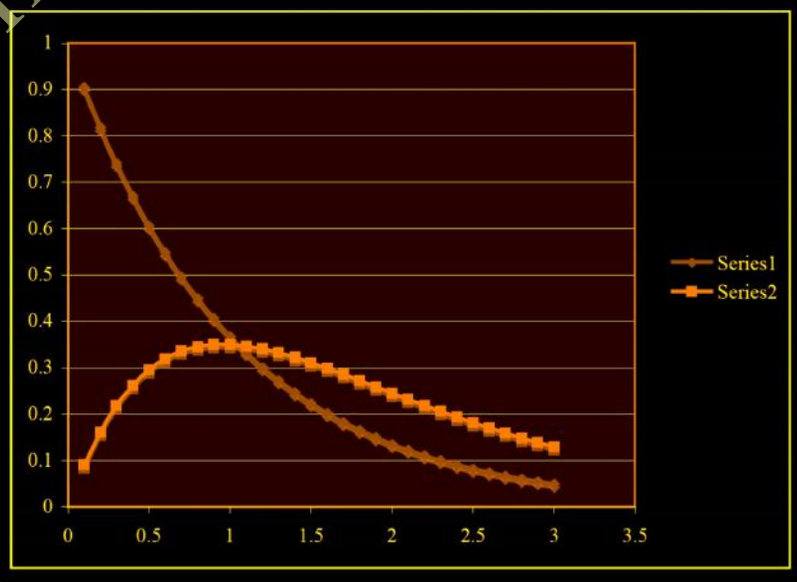
$$\Rightarrow \boxed{N_3 = N_0\left[1 + \frac{\lambda_1 e^{-\lambda_2 t}}{(\lambda_2-\lambda_1)} - \frac{\lambda_2 e^{-\lambda_1 t}}{(\lambda_2-\lambda_1)}\right]} = N_0 - N_2 - N_1$$

Then the time at which concentration of intermediate member (B) will reach maxima is:

$$\left(\frac{dN_2}{dt}\right)_{t=t'} = 0 \Rightarrow \boxed{t' = \frac{\ln\left(\frac{\lambda_2}{\lambda_1}\right)}{\lambda_2-\lambda_1}}$$

A typical graph for the series decay process is shown in adjacent figure for $\lambda_2/\lambda_1 = 1.1$.

→ Here it can be seen that the species B passes through a maximum and then again starts decreasing.



SECULAR EQUILIBRIUM:

Consider equation 2 above, $N_2 = \frac{\lambda_1 N_0}{\lambda_2-\lambda_1}\left[e^{-\lambda_1 t} - e^{-\lambda_2 t}\right]$

Now suppose if $\lambda_1 <<< \lambda_2$ $(\lambda_1 < 10^{-4}\lambda_2)$ or $\tau_1 >> \tau_2$) then after a few half lives of B, $e^{-\lambda_2 t}$ may be assumed to be practically zero and then our equation becomes

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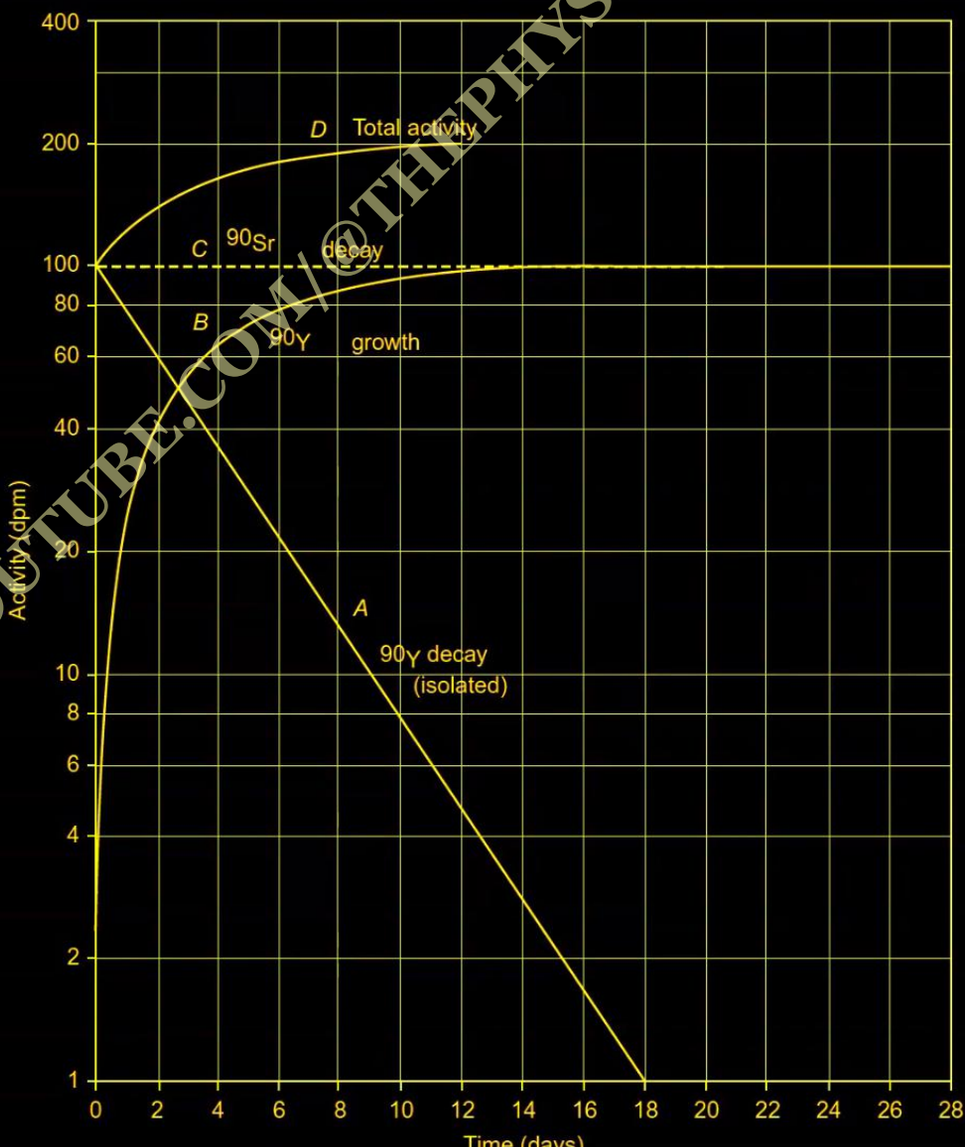
$N_2 \approx \frac{\lambda_1 N_0 e^{-\lambda_1 t}}{\lambda_2 - \lambda_1}$ and since species 1 decays very slowly then for relatively small time periods (few multiples of τ_2), $e^{-\lambda_1 t}$ may be treated as almost 1. Thus for some time we can write,

$$\boxed{N_2 = \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1}}$$

i.e the number of nuclei of species becomes nearly constant for few multiples of τ_2. This phenomenon is known as secular equilibrium. As an example, consider the reaction:

$$^{90}_{38}Sr \xrightarrow{t_{1/2} = 28.8 \text{ years}} {}^{90}_{39}Y \xrightarrow{t_{1/2} = 2.7 \text{ days}} {}^{90}_{40}Zr \text{ (Stable)}.$$

Here after a few days (say about 3 weeks) we expect Yttrium to attain a nearly steady number.



Growth and decay curves following the separation of $^{90}Sr(^{90}Y)$ in secular equilibrium. (A) Decay of isolated ^{90}Y. (B) Ingrowth of ^{90}Y with ^{90}Sr. (C) Decay of isolated ^{90}Sr. (D) Total activity from isolated ^{90}Sr, representing both ^{90}Sr decay and ^{90}Y growth until secular equilibrium is attained. *From*

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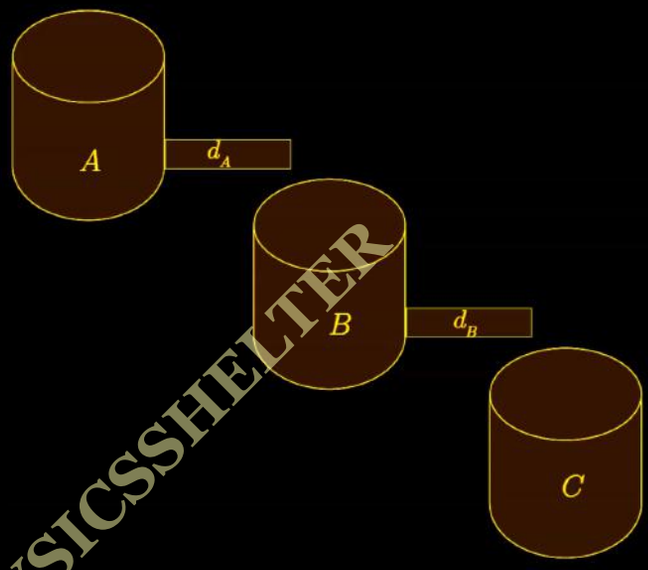
This is clearly visible from the graph drawn above. Here after 14 days, practically the activity has become constant. In secular equilibrium since rate of production of B is same as its decay rate, the rate of accumulation of C is same as the rate of decay of A.

This can be easily understood in terms of an analogy of three barrels. Suppose we are filling barrel C by transferring water from barrel A through barrel B then if diameter of Outlet of A is very small as compared to that of B, the rate of filling of C will practically be independent of diameter of B and B will maintain a level just to balance its outflow and inflow.



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TRANSIENT EQUILIBRIUM:

Consider the equation ② once again $N_2 = \frac{\lambda_1 N_0}{\lambda_2 - \lambda_1}\left[e^{-\lambda_1 t} - e^{-\lambda_2 t}\right]$

Now if $\lambda_2 > \lambda_1$ but nearly of the same order (say $\lambda_1 : \lambda_2 = 1:3$) then, after some time, $e^{-\lambda_2 t} << e^{-\lambda_1 t}$ and we can approximate our equation as:

$$N_2 = \frac{\lambda_1 N_0 e^{-\lambda_1 t}}{\lambda_2 - \lambda_1} = \frac{\lambda_1}{\lambda_2 - \lambda_1} N_1$$

or $\boxed{\frac{N_1}{N_2} = \frac{\lambda_1}{\lambda_2 - \lambda_1}}$ thus here N_1 and N_2 will maintain a nearly constant ratio after a few half lives of N_2. This type of equilibrium is known as transient equilibrium.

NO EQUILIBRIUM:

Consider the reaction $^{56}Ni \xrightarrow{T_H = 6.4\ days} {}^{56}Co \xrightarrow{t_H = 77.3\ days} {}^{56}Fe$

Here practically all the Nickel will decay in a relatively short period and then Cobalt will follow it usual radioactive decay curve after a few half lives of N_i, thus if $\lambda_1 > \lambda_2$ we say there is no equilibrium.

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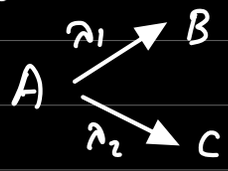
PARALLEL REACTIONS (BRANCHING): Sometimes a species may simultaneously decay in two different modes schematically:



Here for different decay modes we can write, $\left(\frac{dN_A}{dt}\right)_1 = -\lambda_1 N_A$

and $\left(\frac{dN_A}{dt}\right)_2 = -\lambda_2 N_A$ thus net decay rate of A is given

as $\boxed{\frac{dN_A}{dt} = -(\lambda_1+\lambda_2) N_A}$ or for A, $\lambda_{eff} = \lambda_1+\lambda_2$.

Thus we can calculate the count of species A as usual i.e, $N_A = N_{0A}\, e^{-\lambda_{eff} t}$. For species B and C, there will be accumulation. At any time, rates of accumulation of B and C are in the ratio $\lambda_1 : \lambda_2$ thus their total numbers will also be in the same ratio. Thus to find their individual numbers, we can divide the total decayed nuclei in the ratio $\lambda_1 : \lambda_2$.

Accordingly,

$$N_B = N_0\left(1 - e^{-\lambda_{eff} t}\right) \times \lambda_1/(\lambda_1+\lambda_2) \quad \text{and}$$

$$N_C = N_0\left(1 - e^{-\lambda_{eff} t}\right) \times \lambda_2/(\lambda_1+\lambda_2).$$

NUCLEAR FISSION: A fission reaction is an artificially induced nuclear reaction in which a thermal neutron ($KE \approx \frac{3}{2}kT$ at $T \approx 300K$) bombards a heavy nucleus (such as U^{235}) and splits it apart in to neutrons and other nuclides. For example:

$$^{1}_{0}n + {}^{235}_{92}U \longrightarrow {}^{236}_{92}U \longrightarrow {}^{144}_{56}Ba + {}^{89}_{36}Kr + 3\,{}^{1}_{0}n \quad \text{or}$$

$$^{1}_{0}n + {}^{235}_{92}U \longrightarrow {}^{236}_{92}U \longrightarrow {}^{133}_{51}Sb + {}^{99}_{41}Nb + 4\,{}^{1}_{0}n \quad \text{or}$$

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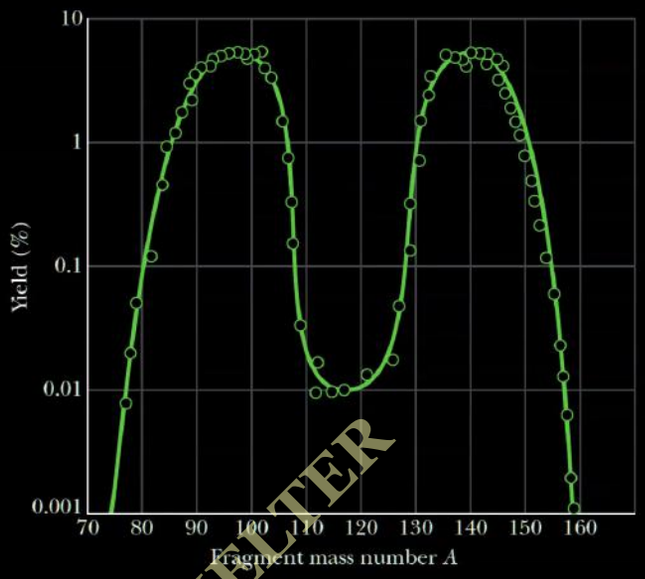
$$ {}^{1}_{0}n + {}^{235}_{92}U \longrightarrow {}^{236}_{92}U \longrightarrow {}^{140}_{54}Xe + {}^{94}_{38}Sn + 2\,{}^{1}_{0}n $$

The splitting process is random, however it has most probability of splitting in to 2:3 mass number ratio. (Adjacent figure shows the probability of splitting vs the mass numbers).

On an average 2.5 neutrons are emitted per fission. The fragment products are usually radioactive and emit β particles in succession to finally achieve stability.



ESTIMATE OF ENERGY RELEASED: The energy released per fission of a heavy nucleus like Uranium is of the order of 200 MeV. This can be estimated as follows. Binding energy per nucleon for a typical heavy nucleus ($A \approx 240$) is about 7.6 MeV and that for medium sized nuclei formed upon fission is about 8.5 MeV per nucleon. Thus the gain in B.E. is 0.9 MeV / nucleon. Thus total gain in Binding energy is $240 \times 0.9 = 216$ MeV

The energy released in nuclear fission comes in the form of Kinetic Energy of the products which can be converted to heat by collisions and interaction with surrounding matter. This heat can then be used to produce electricity (such as in a nuclear reactor) or destroying people and property (such as in Bombs).

Just because a nuclide is heavy, it does not mean that it is readily fissile. For example U^{235} is fissile but U^{238} is not fissile directly (However it can be converted to fissile plutonium in a breeder reactor where it absorbs neutrons and undergoes beta decay to form fissile plutonium). Naturally occuring Uranium is mostly U^{238} and only very small fraction is U^{235}, so concentration techniques have to be used for isolating U^{235} from U^{238}.

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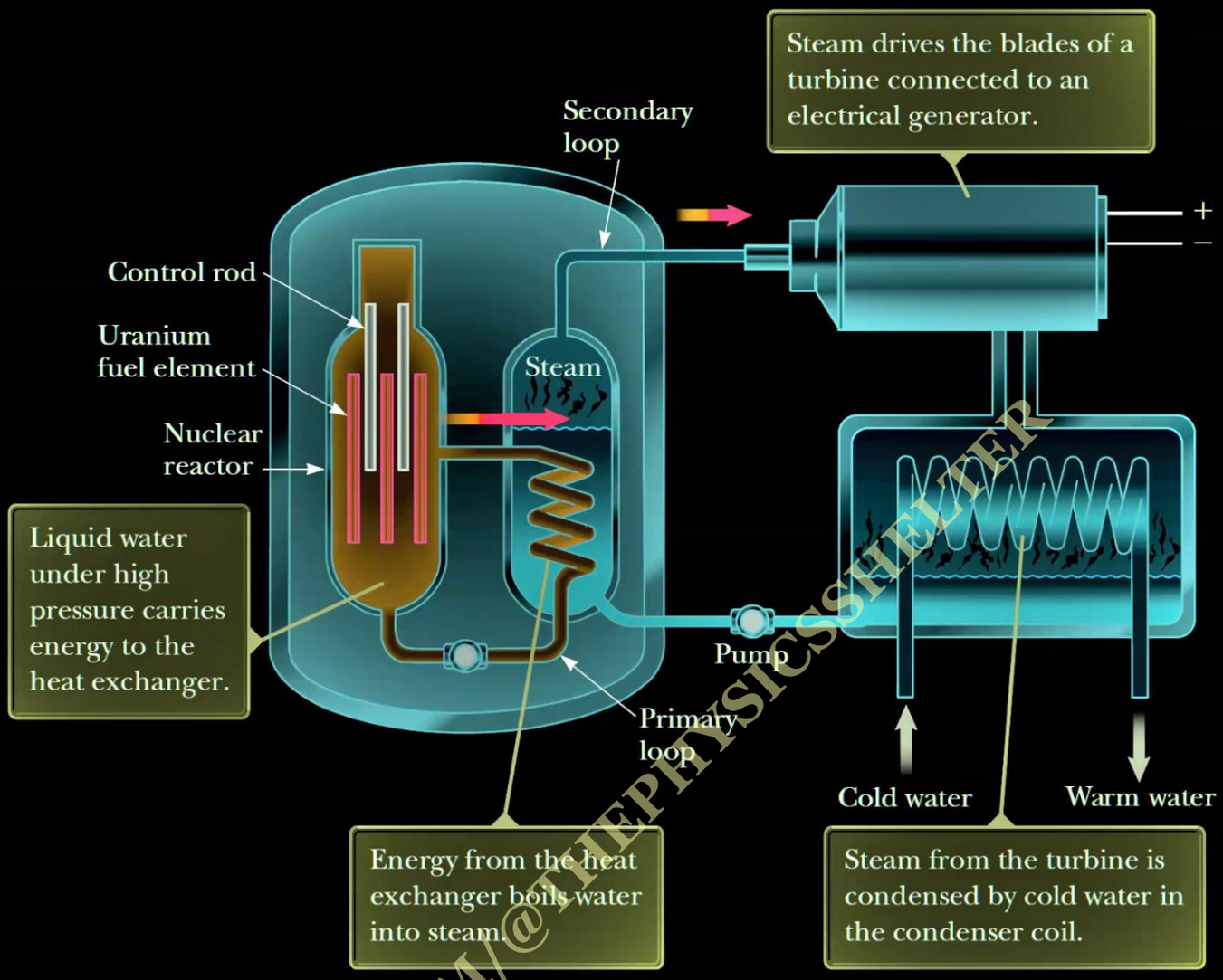
Nuclear Reactor:



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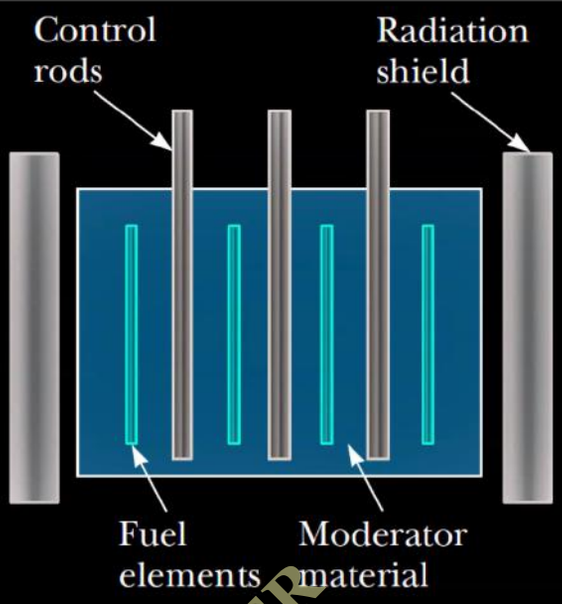
The figure above shows a schematic of a typical fission reactor. The reactor consists of two loops namely primary loop and secondary loop. Primary loop carries energy from the reactor core to the heat exchangers from where, the energy flows in to the secondary loop. The water gets converted to steam in the heat exchanger and the high pressure steam passes through the turbine which in turn drives the coils of a Generator to produce electricity. The student should be familiar with various terms describing a nuclear reactor:

Fuel Rods: Tubes packed with Uranium oxide typically form the fuel rods. A typical reactor contains fuel rods in to assemblies of 200 rods each.

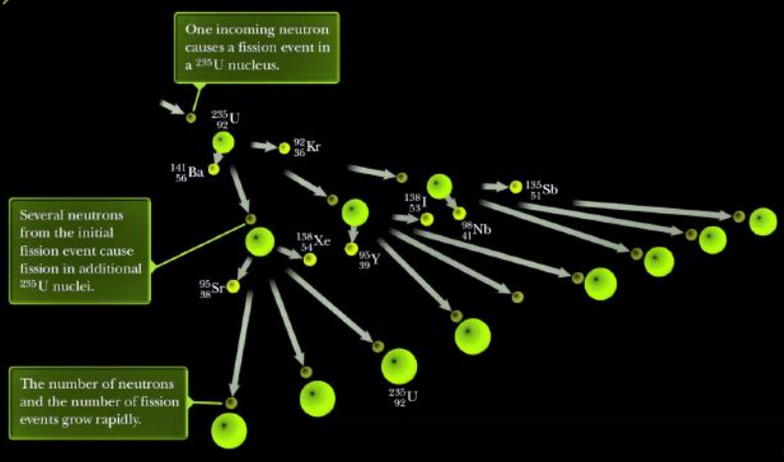
CORE: This is the portion of reactor where controlled fission reaction is carried out. It is typically surrounded with a thick lead walls to absorb any radiation or particles generated in the reactor.



MODERATOR: Fast moving neutrons released in a fission reaction are not capable of causing further reactions. Thus to slow down the neutrons, they are made to collide either with Heavy Water (D_2O) or normal water. This substance fills the Gap between various rods inside the reactor. It also acts as a coolant. (Sometimes Carbon is also used as moderator (Swindne 2011).)

CONTROL RODS: We define a term called multiplication factor (or K factor) as ratio of number of fissions produced by a generation of neutrons to that by the previous generation.

If $K = 1$, the reactor is said to be critical i.e reaction is proceeding at a constant rate. This is desirable for a steady power. If K becomes greater than 1, the reaction rate increases exponentially and can cause hazardous accidents. Such a state is called Supercritical and the K must be quickly brought down close to 1 to quickly absorb excess neutrons. This is the function of control rods. They are typically made of Boron or Cadmium due to their effectiveness in absorbing neutrons. Besides control rods, a reactor core may also have safety rods to quickly extinguish the reaction in case of an emergency.



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COOLANT: Coolant removes energy from the reactor core and supplies it to the steam for power generation (through a heat exchanger).

NUCLEAR HAZARD AND WASTE DISPOSAL: Nuclear waste is hazardous because it emits high energy particles / radiation which can damage the human tissue so it has to be disposed off carefully. Typically nuclear waste is sealed in waterproof containers and buried deep geologic repositories. (Indian atomic programme also tries to convert long lived waste products in to short lived ones before disposal). Besides the usual problem of radioactive waste, Nuclear reactors must be planned very carefully to avoid any accident. In 2011, a Tsunami in Fukushima caused several blasts and melt downs of nuclear reactors. The backup safety measures also failed (Because of power failure, water circulation failed. Generators were supposed to provide backup but flooding of Generator rooms failed the backup). Similarly there were other nuclear disasters like Chernobyl (Ukraine 1986) which killed 30 people and 135000 people had to be evacuated. (Fukushima started due to an earthquake followed by a Tsunami).

FUSION REACTION: When lighter nuclei are bombarded in to each other with sufficiently large energies, they may overcome the coulombic Barrier and fuse together to form a relatively heavier nucleus with greater binding energy per nucleon. Thus fusion of light elements releases energy. Fusion reaction is the main mode of energy Generation in stars. Some examples of such reactions are: (NCERT)

$$ {}_{1}^{1}H + {}_{1}^{1}H \longrightarrow {}_{1}^{2}H + e^{+} + \nu + 0.42\ MeV \quad (1) $$

$$ {}_{1}^{2}H + {}_{1}^{2}H \longrightarrow {}_{2}^{3}He + n + 3.27\ MeV \quad (2) $$

$$ {}_{1}^{2}H + {}_{1}^{2}H \longrightarrow {}_{1}^{3}H + {}_{1}^{1}H + 4.03\ MeV \quad (3) $$

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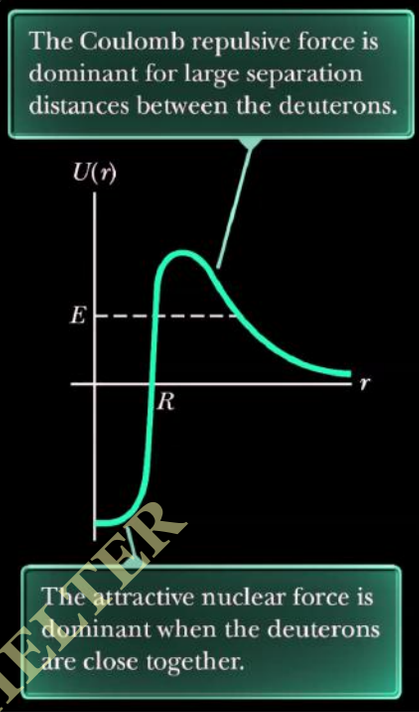
Coulomb Barrier:



For two nuclides to fuse together they have to overcome coulombic barrier. The height of the barrier depends on the charges and radii of interacting nuclei. In the equation ① above, the coulombic barrier between two protons is ~ 400 keV and it will be higher for nuclei with still higher charges. We can estimate the temperature required to initiate this reaction by equating thermal kinetic energy to the coulomb barrier energy i.e $\frac{3}{2}kT = U_{coul} = 400 keV$

which gives us a $T \sim 3 \times 10^9$ Kelvin. When fusion reaction is achieved in this manner by increasing temperature of the sample, we call it thermonuclear fusion. The interior of sun has temperature of only 1.5×10^7 K which is much less than estimated temperature required for such fusion. According to NCERT, such a reaction happens due to protons moving with energies much much greater than average energies. This is also true, however the reaction can also proceed at lower energies through barrier tunneling (similar to the tunneling described in content of alpha decay).

Proton - Proton Cycle (Solar Energy Source):

The fusion reaction in sun is described by following partial reactions:

$$^1_1H + ^1_1H \rightarrow ^2_1H + e^+ + \nu + 0.42 \text{ MeV} \quad (1)$$

$$e^+ + e^- \rightarrow \gamma + \gamma + 1.02 \text{ MeV} \quad (2)$$

$$^2_1H + ^1_1H \rightarrow ^3_2He + \gamma + 5.49 \text{ MeV} \quad (3)$$

$$^3_2He + ^3_2He \rightarrow ^4_2He + ^1_1H + ^1_1H + 12.86 \text{ MeV} \quad (4)$$

Now $2 \times (1) + 2 \times (2) + 2 \times (3) + (4)$ gives:

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$$4\,{}_1^1\mathrm{H} + 2e^- \rightarrow {}_2^4\mathrm{He} + 2\nu + 6\gamma + 26.7\,\mathrm{MeV}$$

or $$(4\,{}_1^1\mathrm{H} + 4e^-) \rightarrow ({}_2^4\mathrm{He} + 2e^-) + 2\nu + 6\gamma + 26.7\,\mathrm{MeV}$$

Thus four hydrogen atoms combine to form an ${}_2^4He$ atom along with a release of 26.7 MeV of energy.

This reaction increases the temperature of the core. If the temperature increases sufficiently (to about 10^8 K), again fusion will occur to cause ${}_2^4He$ nuclei to fuse to form carbon. This type of process can successively generate elements of higher mass numbers but it can't generate elements beyond peak of binding energy curve.

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